

Qubit State Discrimination

with Next-Generation Reservoir Computing

Dissertation

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By

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Abstract

Current quantum computers are prone to errors, requiring error correction protocols that rely on fast and high-fidelity state measurements of quantum bits, or “qubits.” Simultaneous measurement of many qubits can lead to crosstalk, where signals become correlated, complicating state discrimination.

Traditional methods for processing these measurement signals are fast but cannot account for crosstalk and lack nonlinear processing, reducing fidelity. Recently, machine learning (ML) approaches such as neural networks have been used to improve fidelity, but their complexity makes them costly to train, slow to evaluate, and difficult to scale to larger qubit systems.

One alternative is to use a next-generation reservoir computer (NG-RC), an ML model that excels at processing time-series data and is both efficient to train and evaluate. An NG-RC generates linear and nonlinear terms, often polynomial combinations of the input signals at different times, and maps them to the desired output using a weighted sum. The weights are learned through linear regression, which is more efficient than the complex optimization required by neural networks. Additionally, the NG-RC eliminates the need for activation functions like $\tanh$, reducing hardware demands, and can compute polynomials in parallel, reducing latency.

This thesis explores the use of an NG-RC to classify qubit states in two leading quantum computing platforms: superconducting and neutral atom systems.

For superconducting qubit systems, I apply time-averaging and system identification techniques to reduce the number of terms in the model and explore different polynomial degrees for the nonlinear features. I apply this approach to both single-qubit and five-qubit datasets, achieving a 50% (11%) error reduction over traditional methods for the single-qubit (five-qubit) datasets. For the five-qubit chip, this method also achieves 2.5× crosstalk reduction compared to traditional approaches. Model evaluation requires 2 (1) order of magnitude fewer multiplications for single-qubit (multi-qubit) models compared to recent machine learning methods, and uses no nonlinear functions, further improving efficiency. Finally, I discuss how linear regression can be performed

sequentially, enabling real-time model updates to adapt to changing experimental conditions.

For neutral atom systems, I develop two models. The first uses pixel values from fluorescence imaging as the linear terms in the NG-RC, while the second incorporates the average pixel values from neighboring qubits to mitigate crosstalk. I then apply these approaches to a 3×3 array of neutral atom qubits. The crosstalk-mitigating model reduces error by up to 43% and achieves a 3.5 $\times$ reduction in crosstalk compared to traditional methods. The models require four orders of magnitude fewer multiplications than a recent machine learning method, with only a minor 3.5% increase in error across multiple measurement times.

I conclude that NG-RCs provide fast, accurate, and scalable measurements of qubit states across multiple platforms, making them a promising solution for future quantum computers.

Dedication

Dedicated to my parents Gwen and Jason, my brother Will, my sister Emma, and my partner
Isabelle.

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First, I would like to acknowledge the support of Daniel J. Gauthier, who was my advisor for the majority of my time at OSU. His guidance not only helped me become a better writer and scientist, but he also consistently looked out for my personal life, helping me find connections and opportunities beyond this graduate program.

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Publications

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Table of Contents

Abstract	ii
Dedication	iv
Acknowledgements	v
Vita	vii
List of Figures	xii
List of Tables	xv
1 Introduction	1
1.1 Novel Contribution	3
1.2 Outline	7
1.2.1 Chapter 1: Fundamentals of Reservoir Computing	7
1.2.2 Chapter 2: Fundamentals of Quantum Computing	7
1.2.3 Chapter 3: Superconducting Qubit Readout	8
1.2.4 Chapter 4: Superconducting Qubit Readout using Next-Generation Reser- voir Computing	9
1.2.5 Chapter 5: Neutral Atom Readout	9
1.2.6 Chapter 6: Neutral Atom Readout using Next-Generation Reservoir Com- puting	10
1.2.7 Chapter 7: Conclusions and Future Work	11
2 Fundamentals of Reservoir Computing	12
2.1 Machine Learning	12
2.2 Types of Machine Learning Models	13
2.2.1 Feedforward Neural Networks	13
2.2.2 Convolutional Neural Networks	14
2.3 Traditional Reservoir Computing	15
2.4 Next-Generation Reservoir Computing	17

2.5	Summary	18
3	Fundamentals of Quantum Computing	19
3.1	Qubits	19
3.2	Quantum Gates	21
3.3	Quantum Algorithms	23
3.4	Sources of Error	23
3.4.1	Decoherence	23
3.4.2	Measurement Errors	25
3.5	Error Correction Algorithms	25
3.6	Quantum Computing Platforms	26
3.7	Summary	26
4	Superconducting Qubit Readout	27
4.1	Superconducting LC Circuit	27
4.2	The Josephson Junction	29
4.3	The Transmon Qubit	30
4.4	Jaynes-Cummings Model	31
4.5	Dispersive Readout	32
4.6	The Readout Chain	34
4.7	Homodyne Measurement	35
4.8	Frequency Multiplexed Readout	38
4.9	Processing time-dependent I-Q signals	39
4.10	Summary	39
5	Superconducting Qubit Readout using Next-Generation Reservoir Computing	41
5.1	Background	41
5.2	Single Qubit Experimental Setup	42
5.2.1	Multi-Qubit Experimental Setup	42
5.2.2	Traditional approaches to qubit discrimination	43

5.2.3	Machine learning approaches to qubit discrimination	44
5.3	Next-Generation Reservoir Computer Design	47
5.4	Readout Measurement Metrics	48
5.5	Results	49
5.5.1	Single Qubit	49
5.5.2	Multi-Qubit	52
5.6	Discussion	55
6	Neutral Atom Readout	57
6.1	Qubit state encoding	57
6.2	Qubit Register Preparation	58
6.3	Readout	59
6.4	Summary	60
7	Neutral Atom Readout using Next-Generation Reservoir Computing	62
7.1	Background	62
7.1.1	Experimental setup	63
7.1.2	Traditional qubit readout	63
7.1.3	Readout measurement metrics	64
7.2	Enhancing qubit readout using classical machine learning	65
7.2.1	MF-Site	67
7.2.2	MF-Array	68
7.3	Results	69
7.3.1	Classification performance	69
7.3.2	Physical interpretability of weights	73
7.3.3	Algorithm computational complexity	75
7.4	Discussion	77
8	Conclusions and Future Research	79
8.1	Discussion	79

8.2	Future Directions	80
References		88
Appendix A: Methods for Superconducting Qubit Readout		89
8.3	Feature Creation	89
8.4	System Identification	90
8.5	Single-Qubit Model: Training and Testing	90
8.6	Multi-Qubit Model: Training and Testing	91
Appendix B: Methods for Neutral Qubit Readout.		94
8.7	Preprocessing	94
8.8	Estimating Statistical Uncertainty in Infidelity	94
8.9	Training	95
8.10	Optimizing model metaparameters	95
8.11	Testing	95
8.12	Cross-Fidelity Calculation	96

List of Figures

1.1	Schematic of a quantum reservoir processor (QRP) with contributions for each group in the MURI Collaboration.	4
1.2	Diagram of a next-generation reservoir computer.	7
1.3	Bloch sphere with example states. $ \psi\rangle$ has polar angle $\theta = \pi/3$ and azimuthal angle $\phi = 3\pi/4$	8
1.4	Simplified readout diagram for a transmon qubit dispersively coupled to a resonator.	8
1.5	Diagram of next-generation reservoir computer design for classifying superconducting qubit states.	9
1.6	Diagram of readout process for an array of neutral atom qubits. The numbers inside the pixel boundaries represent qubits numbered 1-9.	10
1.7	Matched filter designs for a single qubit with boundary size $s = 6$. (a) MF-Site configuration. (b) MF-Array configuration.	11
2.1	Diagram of a feedforward neural network.	13
2.2	Diagram of a convolutional neural network.	14
2.3	Diagram of a traditional reservoir computer.	16
2.4	Diagram of a next-generation reservoir computer.	17
3.1	Bloch sphere with example states. $ \psi\rangle$ has polar angle $\theta = \pi/3$ and azimuthal angle $\phi = 3\pi/4$	20
3.2	Visualizing single-qubit gates. (a) Pauli-X, (b) Pauli-Y, (c) Pauli-Z, (d) Hadamard, (e) Phase, (f) $\pi/8$	22
4.1	Circuit diagrams of an LC-circuit (a) a Transmon qubit (b) and a tunable Transmon qubit (c).	28
4.2	(top) Transmission of resonator and (bottom) phase shift of transmitted pulse as a function of the incident pulse frequency and qubit state. The parameters for the plots are adapted from [57].	33

4.3	Circuit Diagram of the readout of a Transmon qubit coupled to a resonator, measured in transmission.	34
4.4	Analog I-Q Mixer.	36
4.5	Example I-Q blobs. (a) Example readout measurements for 100K shots. (b) Distribution of states and decision boundary.	37
4.6	Simplified readout diagram for a transmon qubit dispersively coupled to a resonator.	39
5.1	Superconducting qubit readout using the NG-RC algorithm followed by a discriminator.	47
5.2	Classification fidelity and infidelity reduction for single qubit datasets. The matched filter (MF) and temporal postprocessor (TPP) results are taken from Ref. [23]; all other data points are from this work. (a)-(c) Classification fidelity as a function of the number of features in the model for each of the three datasets, where the measurement conditions correspond to the dashed gray lines in (d)-(f), which show the infidelity reduction compared to the MF as a function of the measurement conditions. In the bottom row, the black stars represent the TPP, the blue circles denote the best linear NG-RC, and the red triangles indicate the best nonlinear NG-RC.	50
5.3	Classification fidelity for a 5-qubit chip. Panels (a)–(e) show the classification fidelity for qubits 1 through 5, respectively. The MF and FNN results are taken from Ref. [20], and MF (no boxcar), MF-FNN, MF-RMF-FNN, MF-SVM, and MF-RMF-SVM are taken from Ref. [24]. All other results are from this work.	52
6.1	Diagram of readout process for an array of neutral atom qubits. The numbers inside the pixel boundaries represent qubits numbered 1-9.	59
7.1	Matched filter designs for a single qubit with boundary size $s = 6$. (a) MF-Site configuration. (b) MF-Array configuration.	67

7.2	The measurement infidelity from different analysis methods at different readout times. The square, Gaussian, CNN (site) and CNN (array) infidelity values are taken from Ref. [30]. The error bars represent the standard error of shuffling the dataset with 10 random seeds.	69
7.3	Classification infidelity for each qubit (numbered in top left of each box) for a measurement time of 36 ms. The standard error for the matched filter values is below 0.0001 for each site.	71
7.4	False-3D images of the learned weights for each qubit.	73
7.5	Weights of neighbor qubits (numbered in the top left of each box) for the MF-Array model.	75

List of Tables

1	Summary of my contributions to the qubit state measurement projects.	6
2	Single-Qubit Gates	22
3	Individual qubit readout fidelities and geometric mean fidelity ($\mathbf{F}_{5Q}$) for different designs.	53
4	Mean absolute values of the cross-fidelity at different hamming distances, with $\langle F^{\text{CF}} \rangle_{\Delta k}$ indicating the average over all qubit index separations $\Delta k = k - k' $. . .	54
5	Computational complexity of discriminators.	55
6	Comparison of cross-fidelity values at a measurement time of 36 ms for different methods. The mean standard errors for the MF-Site and MF-Array are 0.0004 for the C-NN values and 0.0002 for the E-E values. The boundary size s is optimized for both the MF-Site and MF-Array.	72
7	Comparison of the number of trainable parameters, evaluation complexity, and fidelity of different approaches. The standard error of the mean fidelity is 0.0003 for both the MF-Site and MF-Array.	76

Chapter 1

Introduction

The fundamental unit of information in a classical computer is a *bit*, which can take on the value 0 or 1. Unlike classical computers, quantum computers use quantum bits, or *qubits*, which can exist in a quantum superposition of two states, $|0\rangle$ and $|1\rangle$, and can become entangled with other qubits, enabling new methods of processing information. Quantum gates are used to manipulate these qubits and perform computations.

There has been growing excitement around quantum computing for a number of reasons. One is its potential to solve problems that are intractable for classical computers. A prominent example is factoring large numbers [1], which could break RSA encryption [2], widely used for secure communication today. They also have the potential to simulate complex quantum systems for drug discovery [3, 4] and to provide more efficient solutions to optimization problems [5–7]. Recent breakthroughs have fueled this excitement, with IBM achieving the first ever 1,000-qubit quantum chip [8], Google demonstrating an exponential suppression of errors [9], and Microsoft’s topological qubits based on a new state of matter [10].

Despite these advancements, we are still in the noisy intermediate-scale quantum (NISQ) [11] era, characterized by a limited number of qubits and error-prone quantum gates and measurements. Many promising algorithms [1, 12, 13] depend on fault-tolerant quantum computers, meaning their error rates are below a critical threshold.

While error correction algorithms [9, 14, 15] offer a partial solution, they rely on frequent mid-circuit measurements of many qubit states. These measurements must be fast to avoid decoherence [16–18], which is the loss of quantum information to the environment, and have high classification fidelity, which is the probability of correctly identifying the qubit states.

For superconducting quantum computers [9, 19], a leading quantum computing platform, qubit states are measured through *readout*, which involves sampling time-series signals from the qubits

and processing them using a discriminator, a classification algorithm that determines their state (e.g. $|0\rangle$ or $|1\rangle$). Multi-qubit devices often use a frequency-multiplexed data bus [20], which can introduce crosstalk through unwanted correlations between qubit signals.

Traditional discriminators for superconducting qubit readout, such as boxcar filters [18] or matched filters [21, 22], are computationally efficient, making them scalable for larger qubit systems. However, they lack nonlinear processing, which can increase fidelity and account for crosstalk.

Machine learning (ML) is a new approach for discriminator design that addresses these problems. One recent study improves upon matched filtering by using linear regression to optimize filter weights [23], but it has not been tested for multi-qubit systems. Other studies use large feed-forward neural networks (FNNs) [20, 24, 25], which can detect crosstalk and improve classification fidelity, but are computationally expensive to train. Also, their complexity increases latency and hardware demands.

Another leading quantum computing platform is based on neutral atom qubits [26–29], where atoms are trapped in an array of optical tweezers. Their states are measured using fluorescence imaging, where each qubit appears as a bright or dark spot on a region of camera pixels, which are processed to determine the qubit states. However, long measurement times are often required to improve the signal-to-noise ratio (SNR), and fluorescence from neighboring qubits can result in crosstalk.

Traditional discriminator algorithms for neutral atom readout involve applying square or Gaussian filters [30] to the pixels corresponding to each qubit. However, these approaches do not account for crosstalk or experimental imperfections such as optical aberrations, reducing fidelity.

Recently, ML approaches, such as convolutional-neural networks (CNNs) [30], have been used to process the pixel values to increase fidelity, but these are computationally expensive to train and evaluate, and their complexity scales poorly as the number of qubits increases.

In this thesis, I introduce new machine learning discriminator algorithms based on next-generation reservoir computing (NG-RC) [31], a model well suited for processing time-series data that is ef-

efficient to train, fast to evaluate, and scalable to larger qubit systems. I apply these algorithms to classify both single and 5-qubit superconducting qubits and a 3×3 array of neutral atoms, comparing their fidelity and computational complexity with traditional methods and recent ML approaches.

In the next section, I summarize my contributions to the MURI collaboration on quantum reservoir processors, as well as my individual contributions presented in this thesis.

1.1 Novel Contribution

While the primary focus of this thesis is improving qubit state readout for quantum computing, the work is inspired by the objectives of a Multidisciplinary University Research Initiatives (MURI) collaboration funded by the Air Force Office of Scientific Research (AFOSR). The overarching goal is to develop superconducting quantum reservoir processors (QRPs) for real-time processing of classical and quantum data. A schematic of the proposed QRP architecture, along with the participating universities and research groups and their respective roles, is shown in Figure 1.1.

The schematic depicts classical data as inputs (input encoding) to a cascade of quantum reservoirs (QRCs), a semi-classical backend (Q-Backend), and a classical backend (C-Backend). The QRC and Q-Backend are designed to leverage quantum effects and nonlinear processing to preprocess the input signals, capturing temporal correlations in the data (see Ref. [32] for more details). The final processing layer is the classical backend (C-Backend), involving the rapid evaluation of a machine learning algorithm using a high-end FPGA.

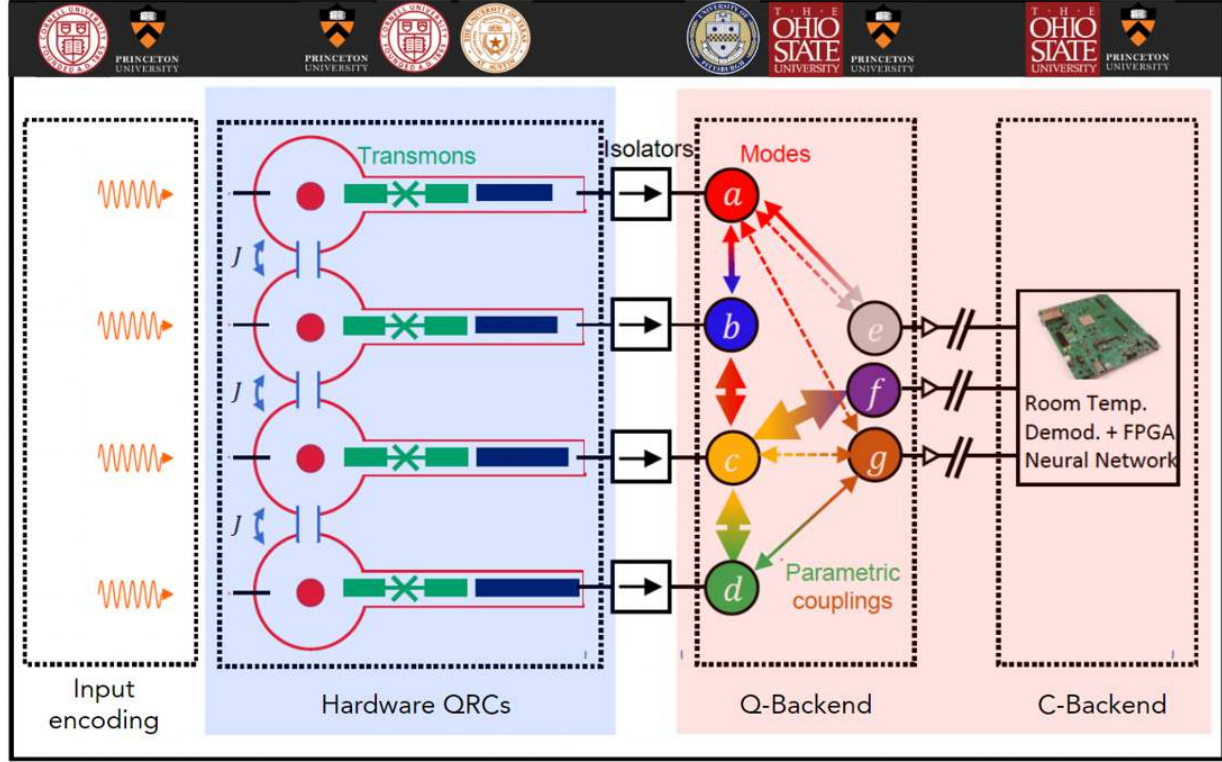


Figure 1.1: Schematic of a quantum reservoir processor (QRP) with contributions for each group in the MURI Collaboration.

The primary role of the QuantInfo group at The Ohio State University is to develop the classical backend. From January 2022 our team consisted of Daniel J. Gauthier (former PI and advisor), Wendson de Sa Barbosa, and myself. Today, the team consists of Gregory Lafyatis (current PI and my advisor) and myself.

Initially, we did not have much experience in rapid evaluation of machine learning models on high-end FPGAs. To build that expertise, we started with a simpler problem: real-time feedback control of a classical dynamical system. We first developed an efficient control algorithm based on an NG-RC, and simulated the control of a chaotic map, documented in Ref. [33]. Next, we implemented our control algorithm on a low-end FPGA to control a chaotic circuit, with an evaluation time of 50 ns, documented in Ref. [34].

Building on this experience, we developed NG-RC algorithms for both superconducting and

neutral atom readout, while also working with high-end FPGAs such as the Xilinx ZCU216 and Xilinx 4x2 boards to understand how to implement and optimize these algorithms in hardware. Although the FPGA implementation and full integration with the quantum processing layers remain future work, the results presented in this thesis lay the foundation for achieving that goal.

In addition to my work on the classical backend, I collaborated with other groups in the MURI project to help develop the quantum hardware for the QRP. This included research visits to the Hatridge group at the University of Pittsburgh, where I performed superconducting qubit characterization with Boris Mesits, and to the Shankar group at the University of Texas at Austin, where I worked with Josiah Cochran on using high-end FPGAs to control quantum systems.

In this thesis, I focus on my work in developing and testing NG-RC algorithms for qubit readout, with my contributions summarized in Table 1. In the next section, I provide an outline of the thesis.

Chapter	Contributions
Chapter 4 Superconducting Qubit Readout using Next-Generation Reservoir Computing	Single-Qubit: • Developed NG-RC models for single qubit readout. • Used time-series averaging and system identification to reduce complexity. • Used nonlinear processing to increase fidelity • Achieved 1-2 orders of magnitude fewer multiplications vs. traditional and ML approaches. • Improved fidelity across all measurement conditions. • Achieved 50% error reduction over traditional approach for three state task. 5-Qubit: • Developed NG-RC models for multiplexed readout. • Achieved higher fidelity than traditional and two machine learning approaches. • Achieved 3 orders of magnitude fewer multiplications and trainable parameters, and 1800 fewer nonlinear activation functions compared to a FNN, with only a 0.7% drop in fidelity. • Demonstrated that batch training can be used to train an NG-RC with large datasets. • Discussed how to perform real-time training.
Chapter 6 Neutral Atom Readout using Next-Generation Reservoir Computing	3×3 Array: • Developed NG-RC models for neutral atom qubits, one with increased crosstalk reduction. • Achieved 43% error reduction compared to traditional approaches • Achieved 2 orders of magnitude fewer trainable parameters, and 4 orders of magnitude fewer multiplications, and no nonlinear activation functions compared to a CNN, with only a minor 3.5% increase in error. • Physical interpretability gives insight into experimental imperfections.

Table 1: Summary of my contributions to the qubit state measurement projects.

1.2 Outline

In this section, I provide a brief summary of each chapter in the thesis, along with key figures.

1.2.1 Chapter 1: Fundamentals of Reservoir Computing

In this chapter, I define machine learning, distinguishing between supervised and unsupervised learning. I then describe some of the machine learning models that serve as benchmarks in later results, including feedforward neural networks (FNNs) and convolutional neural networks (CNNs). Next, I describe traditional and next-generation reservoir computing, concluding that the efficiency in training and evaluation makes an NG-RC, depicted in Figure 1.2, an interesting candidate for qubit state discrimination.

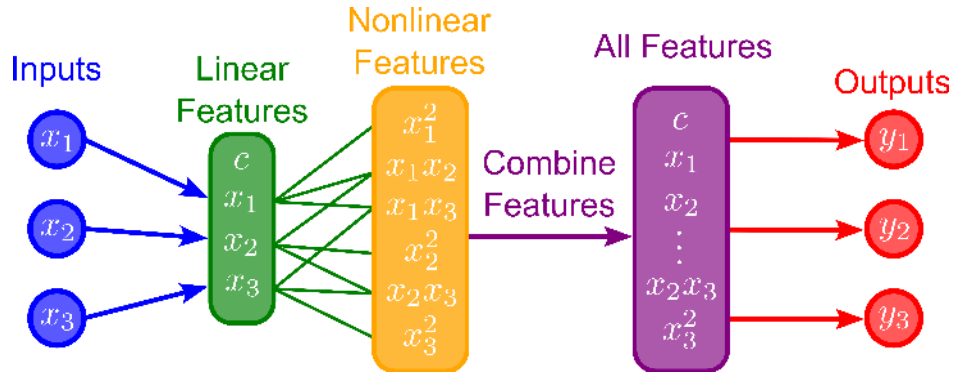


Figure 1.2: Diagram of a next-generation reservoir computer.

1.2.2 Chapter 2: Fundamentals of Quantum Computing

In this chapter, I introduce the basics of qubits, depicted in the Bloch sphere representation in Figure 1.3, and explain why fast and high-fidelity measurements are essential in quantum computing. I focus on challenges such as decoherence and measurement backaction. I also summarize popular quantum computing platforms and motivate my focus on superconducting and neutral atom qubits.

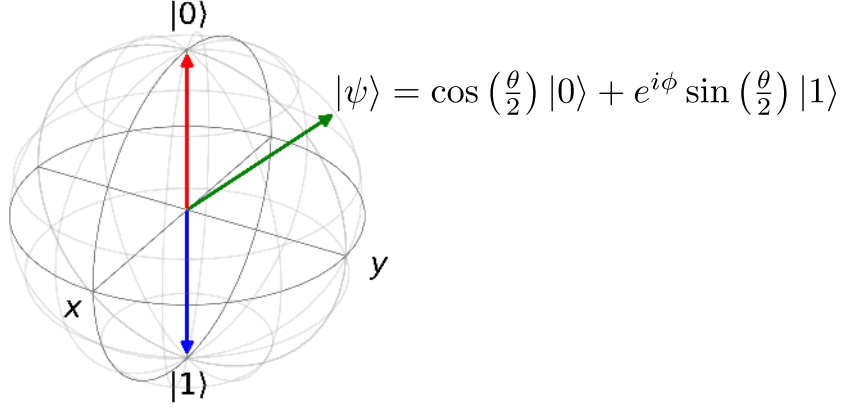


Figure 1.3: Bloch sphere with example states. $|\psi\rangle$ has polar angle $\theta = \pi/3$ and azimuthal angle $\phi = 3\pi/4$.

1.2.3 Chapter 3: Superconducting Qubit Readout

In this chapter, I describe superconducting qubits, including a description of Josephson Junctions and anharmonic quantum oscillators, focusing on a particular type of qubit called a Transmon. Next, I describe the readout process, including the Jaynes-Cummings model, dispersive readout, experimental considerations, analog I-Q mixing, and frequency multiplexed readout. A simplified diagram of the readout scheme is depicted in Figure 1.4.

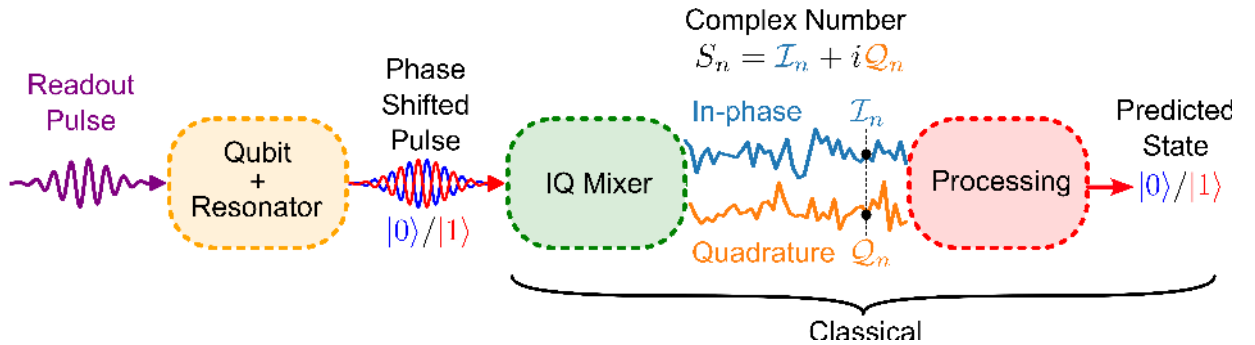


Figure 1.4: Simplified readout diagram for a transmon qubit dispersively coupled to a resonator.

1.2.4 Chapter 4: Superconducting Qubit Readout using Next-Generation Reservoir Computing

In this chapter, I describe the single- and multi-qubit datasets used in this study, traditional and recent machine learning state discriminators, the NG-RC discriminator design (shown in Figure 1.5), and performance metrics. Then, I discuss the results for the single qubit and multi-qubit datasets, comparing the fidelity and computational complexity with traditional and recent machine learning approaches, finishing with a discussion.

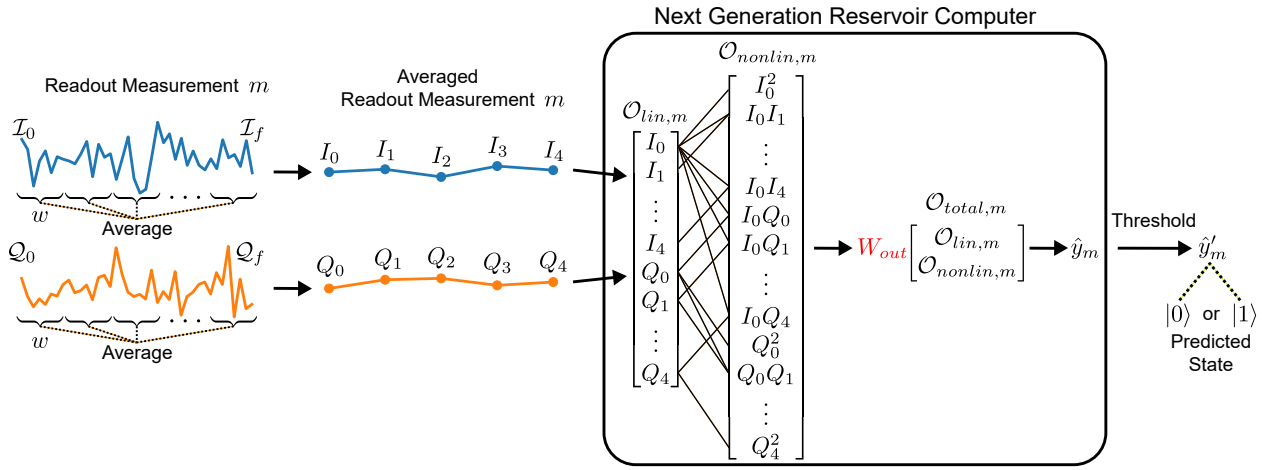


Figure 1.5: Diagram of next-generation reservoir computer design for classifying superconducting qubit states.

1.2.5 Chapter 5: Neutral Atom Readout

In this chapter, I describe how the hyperfine ground states of the neutral atoms are used to encode the qubit states. Next, I briefly describe how the atoms are cooled, trapped, and rearranged using optical tweezers. Finally, I discuss how their states are measured, depicted in Figure 1.6, including challenges such as low SNR, crosstalk, and scalability. I also discuss the difference between destructive and non-destructive measurement approaches.

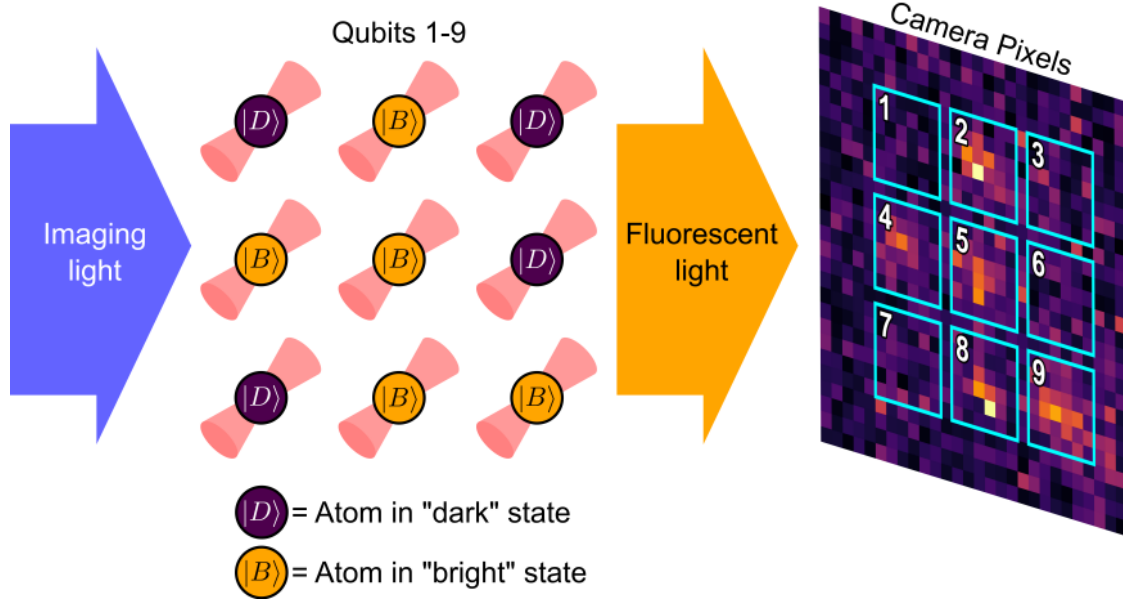


Figure 1.6: Diagram of readout process for an array of neutral atom qubits. The numbers inside the pixel boundaries represent qubits numbered 1-9.

1.2.6 Chapter 6: Neutral Atom Readout using Next-Generation Reservoir Computing

In this chapter, I discuss the experimental setup of the 3×3 array of neutral atoms. I then discuss traditional approaches, performance metrics, and recent machine learning approaches. I then describe the NG-RC models, depicted in Figure 1.7, and compare their fidelity over different measurement times to traditional and machine learning approaches. I then compare crosstalk reduction and complexity with these same approaches. Last, I discuss the physical interpretability of the NG-RC models, and finish with a discussion.

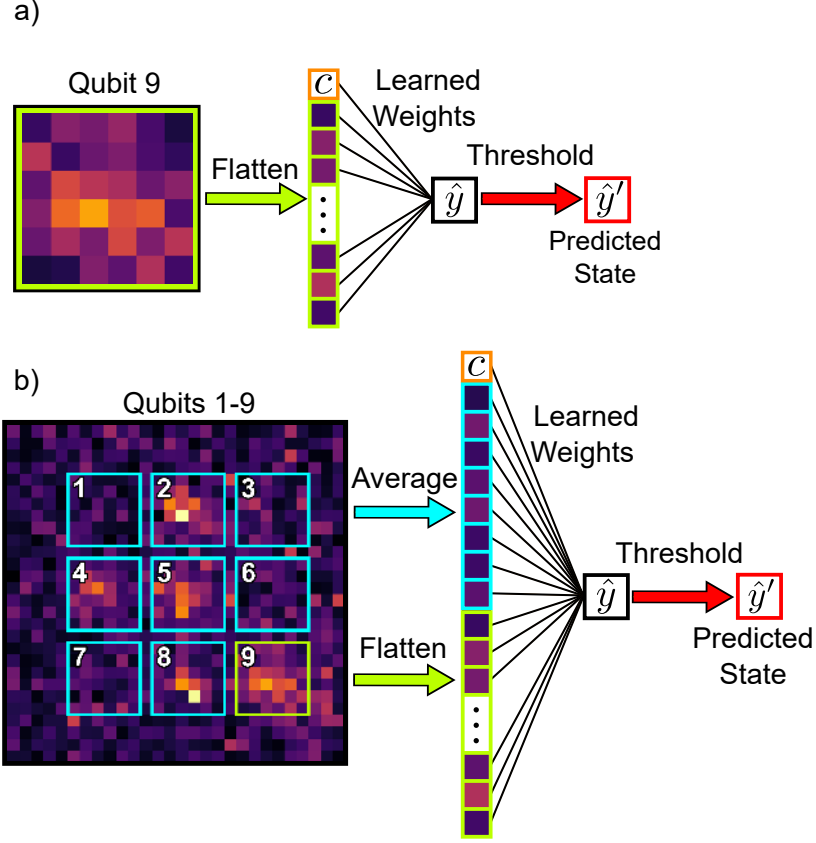


Figure 1.7: Matched filter designs for a single qubit with boundary size $s = 6$. (a) MF-Site configuration. (b) MF-Array configuration.

1.2.7 Chapter 7: Conclusions and Future Work

In this chapter, I summarize the conclusions for the superconducting and neutral atom qubit works, with an emphasis on fidelity, complexity, scalability, and real-time training. I then discuss future possibilities for research, such as scaling to systems with a larger number of qubits and how to implement these models on an FPGA.

Chapter 2

Fundamentals of Reservoir Computing

As this thesis uses a type of machine learning model known as next-generation reservoir computing to discriminate qubit states in different quantum computing platforms, some background on machine learning and reservoir computing is needed to appreciate the contributions of this work.

In this chapter, I provide an overview of machine learning, including supervised learning and common machine learning models used for comparison later in this work. I highlight key properties relevant to this study, particularly the computational complexity of training and evaluating these models. I then introduce reservoir computing, with a focus on next-generation reservoir computing, which is central to this work. Finally, I discuss the computational advantages of next-generation reservoir computing over both conventional machine learning methods and traditional reservoir computing, establishing the motivation for this study.

2.1 Machine Learning

Machine learning (ML) refers to algorithms that learn patterns in data to make predictions or decisions based on new, unseen data. There are two types of machine learning relevant to this work: supervised and unsupervised learning. This work uses supervised learning but later compares results with unsupervised approaches.

Supervised learning involves providing a machine learning model with training data, along with the correct answers (called labels), which represent the desired output for a specific task. The model learns to map the input data to these labels in a process known as “training.” The performance of the model can then be evaluated on unseen data and labels, known as “testing.”

In unsupervised learning, the model is not provided with labels during training. Instead, it identifies patterns, structures, or relationships in the data without explicit guidance.

2.2 Types of Machine Learning Models

There are many types of machine learning models, each suited to different tasks. This section briefly describes two widely used models: feedforward neural networks (FNNs) and convolutional neural networks (CNNs). I highlight these models because they serve as benchmarks for comparison in later results.

2.2.1 Feedforward Neural Networks

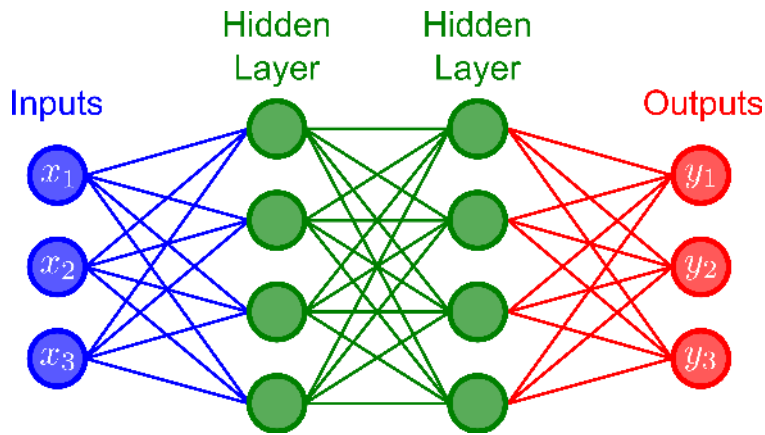


Figure 2.1: Diagram of a feedforward neural network.

A feedforward neural network, depicted in Figure 2.1, is a type of machine learning model where information flows in a unidirectional manner from input to output [35], and typically contains many “hidden layers.” These layers typically consist of many “neurons,” which perform nonlinear transformations of weighted and summed inputs from the previous layer. These nonlinear transformations are typically performed using activation functions such as ReLU, sigmoid, or tanh.

The weights and biases for each neuron are learned through a training process that relies on backpropagation, where errors propagate backward through the network to compute weight updates. These updates are applied using gradient descent, a computationally expensive process involving extensive matrix multiplications and nonlinear function evaluations. Training also requires vast amounts of labeled data to achieve high accuracy, further increasing the computational

burden. To mitigate these demands, specialized hardware such as graphics processing units (GPUs) is often used to accelerate training by performing many of these operations in parallel.

Evaluating the model is also computationally expensive, as it requires the evaluation of many multiplications and nonlinear functions, which are expensive operations to implement in hardware. While GPUs can also be used to accelerate model evaluation, they have inherent latency due to memory transfer bottlenecks that increase their time to make an inference, which can be prohibitive for certain tasks that require ultra-low latency, such as determining the state of a qubit, as discussed in detail later. This necessitates the use of lower latency hardware such as a field-programmable gate array (FPGA) or application-specific integrated circuit (ASIC). However, modern FPGAs have limited computational resources, restricting their ability to evaluate large neural networks.

2.2.2 Convolutional Neural Networks

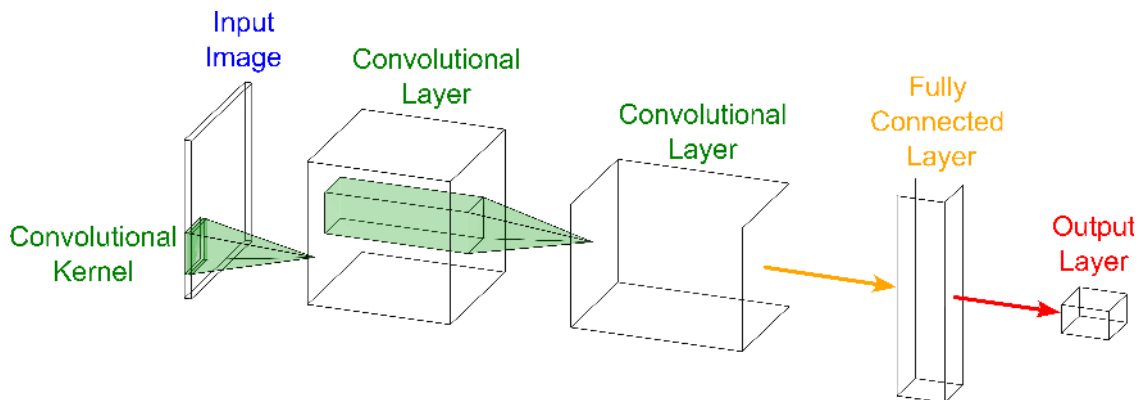


Figure 2.2: Diagram of a convolutional neural network.

Convolutional neural networks (CNNs), depicted in Figure 2.2, are similar to feedforward neural networks but are designed specifically for image processing. The inputs to a CNN are typically the pixel values of an image, where convolutional kernels extract important spatial features by scanning groups of pixels. A convolutional layer applies multiple kernels across the image, generating feature maps that capture different patterns such as edges, textures, and shapes. Pooling layers, not depicted in the figure, commonly follow convolutional layers to downsample these feature maps, preserving key information while reducing dimensionality and computational complexity. Finally,

fully connected layers, similar to the hidden layers in a feedforward neural network, are applied before reaching the output layer.

Like feedforward neural networks, CNNs use backpropagation and gradient descent for training, resulting in high computational complexity. However, CNNs add significant computational overhead due to convolutional operations, which require a large number of multiplications. Evaluating a CNN is also computationally expensive, requiring many convolutional operations and nonlinear activation function evaluations. While GPUs can accelerate both training and evaluation through parallelized operations, the added latency prevents the real-time evaluation needed for certain tasks, such as qubit state discrimination. As with FNNs, specialized hardware such as FPGAs or ASICs may be required for real-time operation, but the size and complexity of CNNs can limit their feasibility on these devices.

As discussed in detail later, the complexity of both FNNs and CNNs do not scale linearly with the number of qubits, which poses a significant challenge for future quantum computers which may have millions of qubits, necessitating the search for more efficient machine learning approaches.

2.3 Traditional Reservoir Computing

A reservoir computer (RC), depicted in Figure 2.3, [36–39] is a type of machine learning model that maps an input into a higher-dimensional space using a fixed “reservoir,” and the readout from the reservoir is trained to produce the desired output using linear regression [40]. The reservoir itself is a “recurrent neural network” (RNN) [38], which unlike a feedforward neural network, has feedback loops in the connection pathways. This allows an RC to retain memory of previous inputs, making it well-suited for learning the temporal behavior of dynamical systems [39, 41].

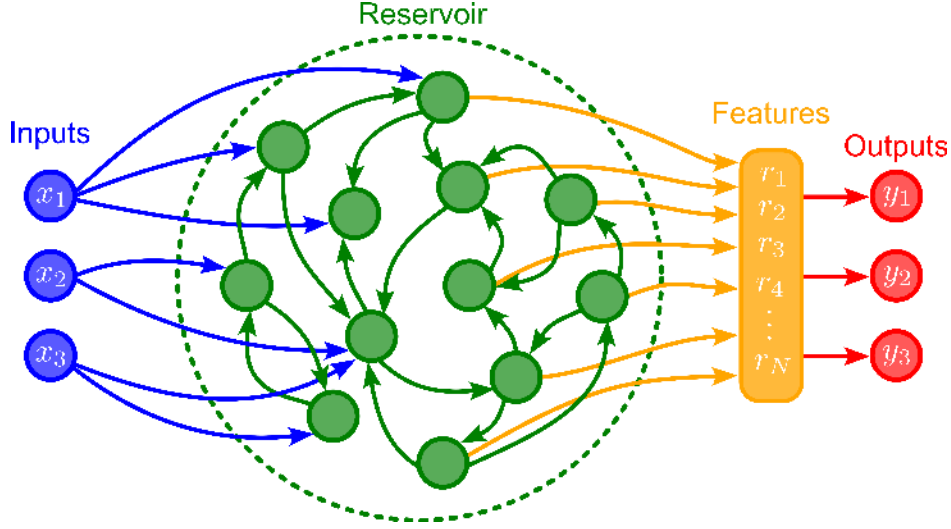


Figure 2.3: Diagram of a traditional reservoir computer.

The reservoir itself can be a physical system such as waves on the surface of a water bucket [42], or implemented in hardware or software with different parameters, such as the number of neurons in the reservoir and how quickly the rate at which memory of past inputs fades. More recently, quantum systems have also been proposed as reservoirs [32, 43, 44].

When processing time dependent data, the state of the reservoir $\mathbf{r}(t)$, comprising all the neurons $\{r_1, r_2, \dots, r_N\}$ in the reservoir, must be updated for each time step. This requires many multiplications and evaluations of nonlinear functions, typically $\tanh$, which make traditional RCs computationally expensive to evaluate in digital hardware. Physical reservoirs have been proposed to address this problem [45], but they suffer from limited reconfigurability.

On the other hand, the training process for an RC is significantly simpler than that of a typical feedforward neural network. This is because the weights connecting the neurons in the reservoir are fixed, requiring no learning. Only the output weights are learned, which represent a linear combination of the reservoir nodes. Crucially, this linear combination only requires regularized linear regression which has a closed-form solution.

There is a newer type of RC called a next-generation reservoir computer (NG-RC) [31] that reduces evaluation complexity while retaining the simple training process of traditional RCs. The primary motivation of this thesis is exploring the NG-RC as a candidate for more efficient qubit

state classification.

2.4 Next-Generation Reservoir Computing

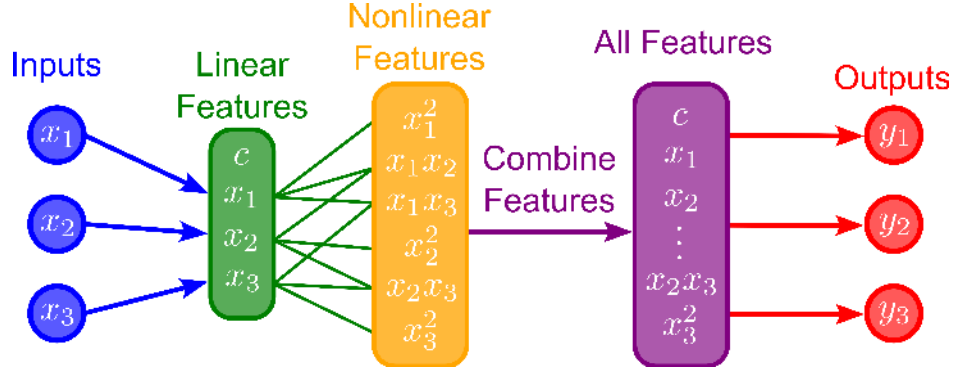


Figure 2.4: Diagram of a next-generation reservoir computer.

A next-generation reservoir computer [31], depicted in Figure 2.4, does not have a “reservoir” with recurrent connections like the traditional approach. Instead, the features typically generated by the reservoir are replaced with linear and nonlinear functions of the inputs at current and previous times, when processing time dependent data. The inputs depicted in the figure have no explicit time dependence, but they could correspond to the same system variable measured at different times (i.e. x_1 and x_2 could correspond to x at t_1 and t_2 , respectively). While the nonlinear features are depicted as polynomial combinations of the inputs, any nonlinear function may be used. The linear and nonlinear features are then concatenated into a total feature vector, depicted in purple, which is analogous to the state of the reservoir in a traditional reservoir computer. Finally, a linear combination of the terms in the total feature vector is applied to get the outputs.

Like a traditional reservoir computer, the output weights are learned through regularized linear regression, making the training process computationally efficient. However, the removal of the reservoir greatly reduces the number of multiplications and eliminates the need for activation functions like $\tanh$, which require additional resources when evaluated on an FPGA. It also has a fewer number of parameters to tune relative to a traditional reservoir computer, with the primary parameters being the polynomial order and the number of previous time steps of the inputs to

include in the linear features.

In a previous work based on using an NG-RC in a control algorithm [34], I showed that the complexity of the model can be reduced further using system identification [46, 47], which selects only the most important polynomial features to include in the model. This technique is also used in later results for the qubit state discrimination task.

2.5 Summary

In summary, both feedforward neural networks and convolutional neural networks are computationally expensive to train and evaluate. A traditional reservoir computer is the best type of machine learning model for processing time dependent data, and training is much more efficient than other approaches, but it is still computationally expensive to evaluate. A next-generation reservoir computer is both simple to train and evaluate, and retains the processing power for time dependent data.

These qualities are essential to the task of qubit state discrimination, as ultra-low latency and scalability with increasing number of qubits is essential. Additionally, for certain quantum computing platforms, such as those that use superconducting qubits, the qubit states are determined by processing time-series data, which is a task that a reservoir computer excels at. All of these qualities motivate the exploration of using an next-generation reservoir computer for more efficient qubit state discrimination in this thesis.

The next chapter provides the fundamentals of quantum computing, focusing specifically on what makes fast and accurate qubit state discrimination crucial to the success of quantum computers, necessitating the search for more efficient machine learning models for this task, such as a next-generation reservoir computer.

Chapter 3

Fundamentals of Quantum Computing

As this thesis focuses on applying novel machine learning techniques to qubit state discrimination to address challenges faced by current quantum computers, some background in quantum computing is necessary to fully appreciate its contributions. While many resources, such as Ref. [48] provide a comprehensive background, this chapter focuses on the key concepts necessary to understand this work.

This chapter is organized as follows: I describe the basics of qubits, quantum gates, and quantum algorithms. Next, I discuss some sources of error that plague quantum computers, such as decoherence and measurement errors. Last, I briefly discuss the role of measurement in error correction. Throughout, I highlight the critical role of fast and accurate qubit state measurements in advancing quantum computing.

3.1 Qubits

The fundamental unit of information in a classical digital computer is a “bit,” which can take on values 0 or 1. In contrast, a quantum bit, or “qubit” is used to represent information in a quantum computer. We represent the computational basis states of a qubit as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (1)$$

Unlike classical bits, qubits can exist in a superposition of these basis states

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (2)$$

where the coefficients α and β are complex numbers, known as probability amplitudes, that determine the likelihood that the qubit is measured in the $|0\rangle$ or $|1\rangle$ state respectively, and must satisfy $|\alpha|^2 + |\beta|^2 = 1$. The state of a qubit can be visualized as a point on the surface of the “Bloch sphere” [48], shown in Figure 3.1 for the two computational basis states and a superposition state.

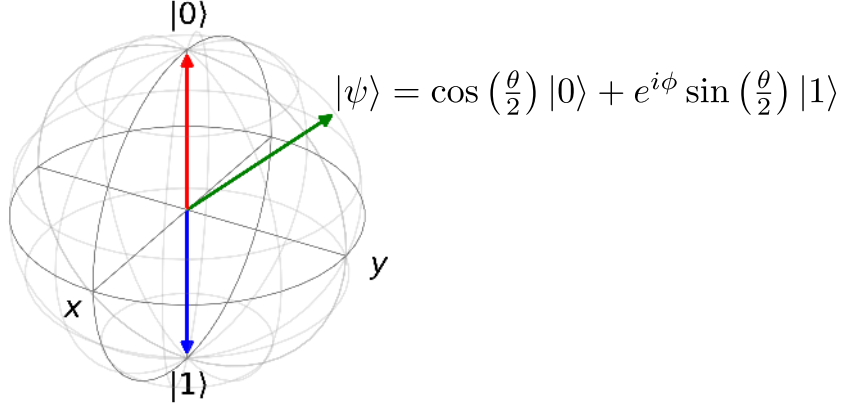


Figure 3.1: Bloch sphere with example states. $|\psi\rangle$ has polar angle $\theta = \pi/3$ and azimuthal angle $\phi = 3\pi/4$.

Although a qubit’s state can exist in any point on the Bloch sphere, a measurement in the computational basis necessarily collapses the wave function into $|0\rangle$ or $|1\rangle$, converting quantum information to classical information. However, quantum computers require these measurements to be highly accurate, performed before quantum information is lost, and capable of preserving the qubit for further computations. These challenges are discussed in detail in Section 3.4 and are a key focus of this thesis.

In quantum computing, it is often necessary to represent the states of multiple qubits as a single, multi-qubit state. If we have two qubits in the states $|\psi_1\rangle = \alpha_1 |0\rangle + \beta_1 |1\rangle$ and $|\psi_2\rangle = \alpha_2 |0\rangle + \beta_2 |1\rangle$, we can represent their combined state using the tensor product

$$|\psi_{12}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle, \quad (3)$$

where $\otimes$ represents the tensor operation. We can then expand the tensor product to get

$$|\psi_{12}\rangle = \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle, \quad (4)$$

where $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ are the computational basis states of the two-qubit system.

In general, an n -qubit system can exist in a superposition of 2^n basis states, while a classical n -bit system can only represent one of 2^n possible values at a time. This exponential scaling of the state space is one of the primary reasons that quantum computers have an advantage over classical computers for certain tasks.

The other key property that provides quantum computers with an advantage over classical computers is entanglement, which is when the states of two or more quantum systems cannot be described independently of each other. In other words, they cannot be separated into tensor products like those in Equation 3. Entanglement is essential to certain quantum algorithms and error correction, and is described in greater detail in [48].

3.2 Quantum Gates

Similar to how logic gates are used to manipulate bits in classical computing, quantum gates are used to manipulate the states of qubits. In the case of a single qubit, these gates correspond to rotations on the Bloch sphere. Importantly, quantum gates are unitary operators, which mean they conserve probability and are reversible, which is not true for all classical gates.

The most common single-qubit gates along with their symbols are listed in Table 2, and they can be visualized on the Bloch sphere in Figure 3.2. The Pauli-X gate, also known as a π -pulse, performs a π rotation around the x -axis on the Bloch sphere. This is the quantum equivalent of the NOT gate, mapping $|0\rangle \rightarrow |1\rangle$ and $|1\rangle \rightarrow |0\rangle$, and is later used in this thesis to initialize qubit states.

Gate Name	Gate Symbol	Matrix
Pauli-X (X)	$\text{---}\boxed{X}\text{---}$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)	$\text{---}\boxed{Y}\text{---}$	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)	$\text{---}\boxed{Z}\text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)	$\text{---}\boxed{H}\text{---}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)	$\text{---}\boxed{S}\text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)	$\text{---}\boxed{T}\text{---}$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

Table 2: Single-Qubit Gates

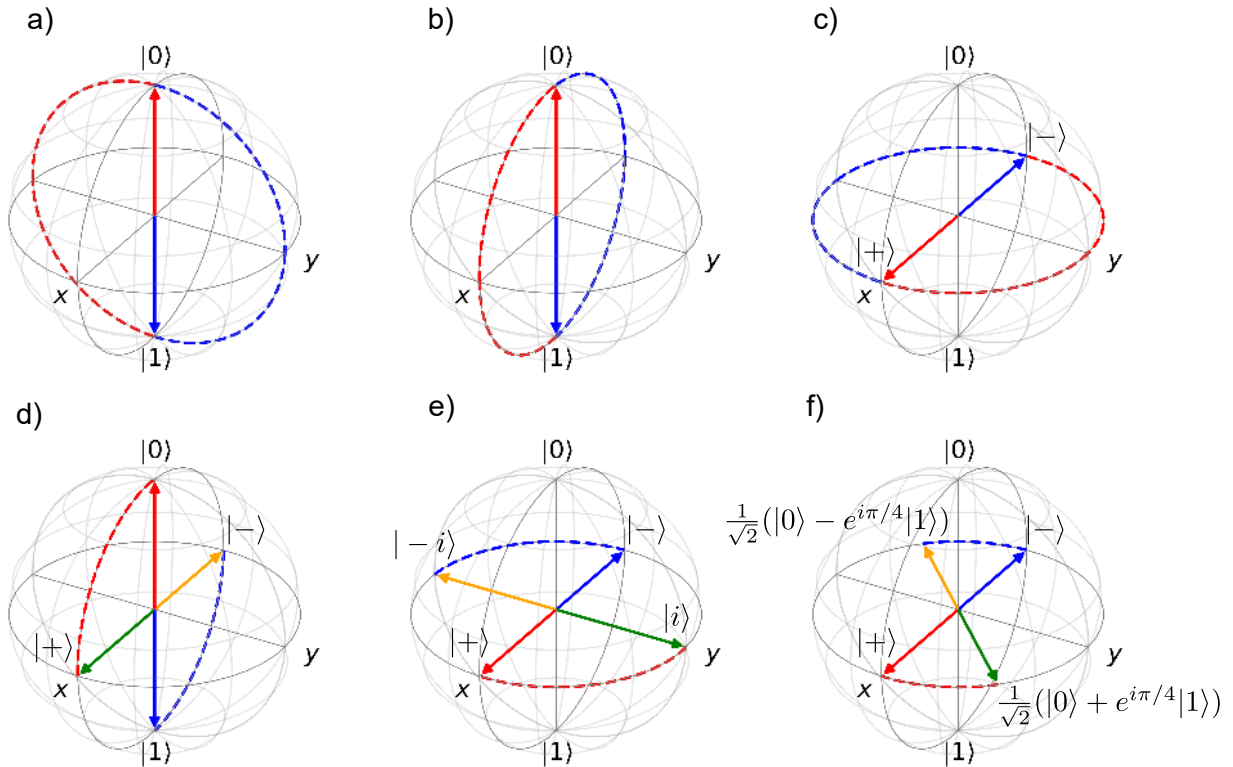


Figure 3.2: Visualizing single-qubit gates. (a) Pauli-X, (b) Pauli-Y, (c) Pauli-Z, (d) Hadamard, (e) Phase, (f) $\pi/8$

Multi-qubit gates enable interactions between qubits, including entanglement, and can be used in succession to form quantum circuits which serve as the building blocks of quantum algorithms. Useful multi-qubit gates can be found in [48].

3.3 Quantum Algorithms

As mentioned in previous sections, only a few tasks are currently known where quantum computers may offer an advantage over classical computers. Notable examples include Shor’s algorithm [1], which provides a superpolynomial speedup for integer factorization; Grover’s algorithm [13], which offers a quadratic speedup for unstructured search problems; and the Quantum Approximate Optimization Algorithm (QAOA) [5], which can provide a more efficient approach to solving certain optimization problems than classical methods. However, these algorithms have yet to realize their full potential, as they require larger numbers of qubits with lower error rates.

3.4 Sources of Error

Though certain quantum algorithms have the potential to outperform classical computers, most have not yet been realized due to the high error rates in current quantum hardware. Quantum systems are affected by multiple sources of noise, fabrication defects that cause qubit inconsistencies, and imperfect quantum gates. While error correction schemes provide part of the solution, they require quantum computers with sufficiently low error rates to function effectively. This section summarizes different sources of error that introduce challenges to the measurement process across different quantum computing platforms.

3.4.1 Decoherence

One of the primary difficulties that quantum computers face is decoherence, which is when information about a quantum state is lost due to unwanted interactions with the environment. Despite efforts to minimize these interactions, they are still present to some degree in all quantum computing platforms. The timescale of decoherence varies by platform, placing time constraints on how

quickly quantum gates and measurements must be executed. There are two types of decoherence that are central to quantum computing, characterized as longitudinal relaxation and transverse relaxation, also known as dephasing. A comprehensive discussion of decoherence can be found in [18].

Longitudinal Relaxation occurs when a qubit loses energy to the environment, causing it to relax from the excited state $|1\rangle$ to the ground state $|0\rangle$. The timescale over which relaxation occurs is the relaxation time T_1 , also known as the qubit’s lifetime, which characterizes the exponential decay of the excited state population $P_1(t) = e^{-t/T_1}$. As an example, in Google’s recent Willow quantum processor [9] which uses superconducting qubits (discussed in the next chapter), the mean relaxation time is $68\mu s$. Relaxation is one possible source of bit-flip errors ($|1\rangle \rightarrow |0\rangle$).

Dephasing, also known as loss of phase coherence, occurs when information about the phase of the qubit ϕ is lost, which corresponds to its rotation around the z -axis on the Bloch sphere. The rate of dephasing is characterised by the dephasing time T_2 or the inhomogeneous dephasing time T_2^* , depending on the measurement method. Following the Bloch-Redfield model of decoherence [17], $1/T_2 = 1/(2T_1) + 1/T_\phi$, where T_1 captures energy relaxation effects and T_ϕ is the pure dephasing time.

Regardless of the platform, a primary challenge in quantum computing is performing all operations and measurements necessary for executing a quantum algorithm on a timescale much shorter than these decoherence times. While error correction schemes can detect and correct the bit-flips or phase-flips caused by these processes, they also rely on large number of operations and measurements. To suppress the growth of errors, both operations and measurements must be performed quickly and accurately. For example, in a superconducting computer the measurement time is on the order of a few hundred nanoseconds, which is a non-negligible fraction of the decoherence time. The use of machine learning algorithms to improve measurement accuracy introduces additional latency, further increasing the risk of decoherence. Minimizing this additional latency is a key focus of this thesis.

3.4.2 Measurement Errors

Measurement errors arise from several factors, including short measurement times (which reduce the risk of decoherence but can compromise accuracy), noisy measurement signals, and state transitions that may occur mid-measurement. Additionally, some errors occur due to crosstalk, which often manifests as interference between measurement signals from multiple qubits.

Measurement backaction can introduce errors when a measurement unintentionally alters the qubit state, such as bit-flips or transitions to higher excited states. This is a key factor in determining whether a measurement is quantum nondemolition (QND). A QND measurement ensures that once a qubit’s state is measured, it remains in that state during subsequent evolution, allowing for repeated measurements without disturbing the quantum information. QND measurements are essential to the success of quantum computers, as discussed later.

The accuracy of measurement is quantified by the measurement fidelity, which is determined by the ratio of correct measurements to the total number of measurements. For superconducting qubits, neutral atom qubits, and trapped ion qubits, recent works report measurement fidelities of 99.9% [49], 99.1% [50], and 99.88% [51], respectively. While these fidelity values appear high, errors can rapidly accumulate due to the large number of measurements required for executing quantum algorithms and implementing error correction schemes.

3.5 Error Correction Algorithms

As described above, errors are common across all quantum computing architectures, making the detection and correction of those errors paramount to enable quantum computers to evaluate complex quantum algorithms. Many of these algorithms require a fault-tolerant quantum computer, meaning the error rates must be reduced below a critical error threshold, beyond which errors accumulate faster than they can be corrected.

Error correction algorithms typically involve encoding a “logical qubit” into multiple “physical qubits” [14], with the physical qubits often arranged in a 2D lattice [9]. These codes detect and

correct errors in the physical qubits by measuring stabilizers, which are specific measurements that can detect bit- or phase-flip errors while preserving the logical qubit’s state. Crucially, these algorithms all require fast and accurate measurements of many qubit states, which is a central focus of this thesis.

3.6 Quantum Computing Platforms

There are many different quantum computing platforms that are currently under active development, including superconducting qubits [9], trapped ions [52], neutral atoms [29], photonic qubits [53], and silicon-based qubits [54] to name a few. Each of these platforms have advantages and drawbacks, which can be understood through the DiVincenzo Criteria [55]. These criteria define the fundamental requirements for a viable quantum computing platform, including scalability, reliable qubit initialization, relatively long coherence times compared to gate and measurement durations, a universal set of quantum gates, and the ability to measure individual qubit states. In this thesis, I focus on two leading quantum computing platforms, superconducting quantum computing and neutral atom quantum computing, as they show promise in achieving the above criteria.

3.7 Summary

This chapter covered the fundamentals of quantum computing, highlighting that the rapid and high-fidelity measurement of many qubit states is essential to the success of future quantum computers. In the next chapter, I use the framework discussed in this chapter to describe the measurement process for one of the leading quantum computing platforms—superconducting qubits.

Chapter 4

Superconducting Qubit Readout

Superconducting qubits are artificial atoms composed of superconducting circuit elements. These qubits exhibit discrete energy levels, with the basis states $|0\rangle$ and $|1\rangle$ encoded in the ground and first excited states, respectively. Quantum gates are implemented by applying microwave pulses to the qubits. This platform offers several advantages, including rapid, high-fidelity gate operations, scalability, compatibility with existing microwave technology, and fabrication using preexisting techniques from the semiconductor industry. However, one primary drawback is the short coherence times of superconducting qubits, making fast and high-fidelity measurements crucial to the success of this platform.

The theoretical framework describing these phenomena falls under circuit quantum electrodynamics (cQED), a well-established field with extensive literature covering its principles in great detail [18, 56–59]. Here, I focus only on the essential aspects of cQED that are directly relevant to the contributions of this thesis and refer the reader to these works for a more comprehensive treatment of the subject.

In this chapter, I provide background on superconducting qubits, including their relation to the quantum harmonic oscillator and the role of Josephson junctions. Then, I describe the process of measuring the state of a superconducting qubit, known as “readout.”

4.1 Superconducting LC Circuit

To understand the basics of superconducting qubits, it is helpful to start with inductor-capacitor (LC) circuits, the direct electrical analog of a classical harmonic oscillator. Such a circuit consists of an inductor with inductance L and a capacitor with capacitance C in a closed loop. When the capacitor is charged and then allowed to discharge through the inductor, energy oscillates between the capacitor’s electric field and the inductor’s magnetic field, causing a continuous exchange of

charge Q on the capacitor and magnetic flux Φ in the inductor. This oscillatory behavior is governed by the natural resonance frequency $\omega_r = 1/\sqrt{LC}$. A diagram of an LC circuit is shown in Figure 4.1a.

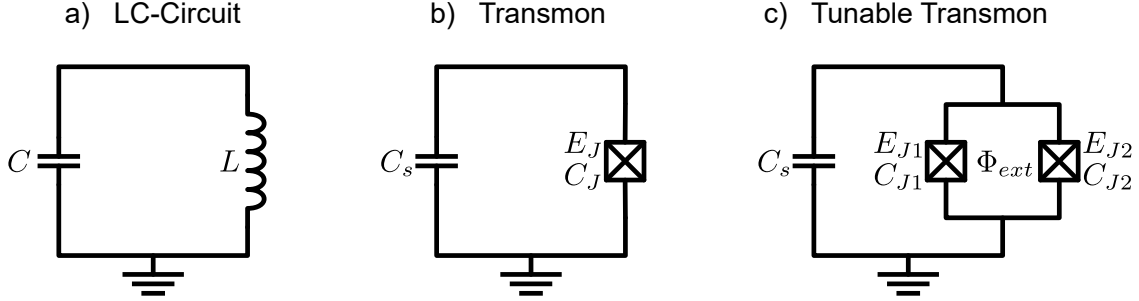


Figure 4.1: Circuit diagrams of an LC-circuit (a) a Transmon qubit (b) and a tunable Transmon qubit (c).

Even without a resistor, small resistances in the wires and other components result in energy dissipation over time. However, if the circuit is made using materials such as aluminum or niobium and cooled below a critical temperature, it becomes superconducting, eliminating all electrical resistance. This occurs because at sufficiently low temperatures, lattice vibrations (phonons) mediate an attractive interaction between electrons, allowing them to form Cooper pairs despite their natural Coulomb repulsion. While individual electrons are fermions and follow the Pauli exclusion principle, a Cooper pair behaves as a boson, meaning multiple pairs can occupy the same quantum state. As a result, these pairs condense into a macroscopic quantum state described by a single wavefunction, as explained by BCS theory [60].

As the charge is now carried by Cooper pairs which follow quantum mechanical rules, the charge Q on the capacitor and magnetic flux Φ on the inductor cannot be simultaneously measured with arbitrary precision, resulting in the commutation relation $[\hat{\Phi}, \hat{Q}] = i\hbar$. This constraint makes the superconducting LC circuit behave as a quantum harmonic oscillator described by the Hamiltonian

$$\hat{H} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L} \quad (5)$$

where $\hat{Q}$ and $\hat{\Phi}$ are now quantum operators. Following [18], we define the reduced flux (also

known as the gauge-invariant phase across the inductor) as $\hat{\phi} \equiv 2\pi\hat{\Phi}/\Phi_0$, where $\Phi_0 = h/(2e)$ is the superconducting magnetic flux quantum. Similarly, we define the reduced charge as $\hat{n} = \hat{Q}/(2e)$, which represents the number of Cooper pairs that have tunneled to the “island”—the region between the inductor and the capacitor. This results in the new Hamiltonian

$$H = 4E_C\hat{n}^2 + \frac{1}{2}E_L\hat{\phi}^2 \quad (6)$$

where $E_C = e^2/(2C)$ is the charging energy required to add each electron to the island and $E_L = (\Phi_0/2\pi)^2/L$ is the inductive energy. Equivalently, the Hamiltonian can be written in the classic form as

$$H = \hbar\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right), \quad (7)$$

where $\hat{a}^\dagger$ and $\hat{a}$ are the raising and lowering operators, respectively, with energy eigenvalues given by

$$E_k = \left(k + \frac{1}{2} \right) \hbar\omega \quad (8)$$

where k is the quantum number that describes the energy eigenstates of the system. It is important to note that the energy levels are evenly spaced by $\hbar\omega$, which makes it challenging to isolate only two eigenstates to encode the qubit state. This is because any applied excitation intended to drive the system from the first excited state to the second would also drive transitions to higher excited states. One solution is to introduce anharmonicity to the system, resulting in unevenly spaced energy levels.

4.2 The Josephson Junction

The anharmonicity required to create a superconducting qubit can be achieved through a nonlinear inductor in the form of a Josephson junction [61], which consists of a thin insulating barrier between two superconducting materials. Cooper pairs can tunnel through this barrier without resistance, a phenomenon known as the Josephson effect. This modification allows Equation 6 to be

rewritten as

$$H = 4E_C \hat{n}^2 - E_J \cos(\hat{\phi}), \quad (9)$$

where $E_c = e^2/(2C_\Sigma)$, $C_\Sigma = C_s + C_J$, C_s is the capacitance of the “shunt capacitor” and C_J is the self-capacitance of the Josephson junction. We see that the nonlinear inductance has the effect of modifying the potential from $\sim \hat{\phi}^2$ to $\sim \cos(\hat{\phi})$. Expanding the nonlinear potential, we see that

$$E_J \cos(\hat{\phi}) = \frac{1}{2}E_J \hat{\phi}^2 - \frac{1}{24}E_J \hat{\phi}^4 + \mathcal{O}(\hat{\phi}^6), \quad (10)$$

where the leading term corresponds to a quantum harmonic oscillator potential, similar to the linear inductance case, while the second term introduces a weak anharmonicity, resulting in uneven energy level spacings suitable for a qubit. This modified circuit is pictured in Figure 4.1b, where the Josephson Junction is symbolized by a box with an “X” through it.

4.3 The Transmon Qubit

The parameters and configuration of Josephson Junctions in these superconducting circuit designs can be used to realize different types of qubits, each with distinct benefits and drawbacks. One example is the Transmon qubit, characterized by $E_J \gg E_C$ (pictured in Figure 4.1b). A common modification is to include a second junction in parallel with the first, forming a superconducting quantum interference device (SQUID) that allows the effective E_J to be tuned through an external magnetic flux Φ_{ext} through the loop, adjusting the qubit’s frequency. This tunable transmon is pictured in Figure 4.1.

The results on superconducting qubits in this thesis use Transmon qubits as they are a popular choice in the industry, with both IBM [19] and Google [9] using variations of the Transmon qubit in their design. However, the measurement process for different types of superconducting qubits is similar, making later results generalizable to other superconducting qubit platforms.

4.4 Jaynes-Cummings Model

As mentioned in Section 3.4, quantum computers not only require fast and accurate measurements of qubit states, but these measurements must also be quantum nondemolition (QND) to avoid altering the state of the qubits in an unexpected manner.

For superconducting qubits, QND measurements are achieved through dispersive readout, in which a resonator is coupled to the qubit (see Figure 4.3), serving as a measurement probe. Rather than directly measuring the qubit's state, a classical measurement is performed on the resonator, which encodes information about the qubit state. In the dispersive coupling regime, the qubit and resonator do not exchange energy directly, minimizing measurement backaction and preserving the qubit's state. This interaction between the qubit and the resonator is described by the Jaynes-Cummings Hamiltonian [62], described below

$$H_{\text{JC}} = \underbrace{\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)}_{H_r} + \underbrace{\frac{\omega_q}{2} \hat{\sigma}_z}_{H_q} + \underbrace{g (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger)}_{H_{\text{int}}}, \quad (11)$$

where H_r , H_q , and H_{int} represent the Hamiltonians of the resonator, qubit, and their interaction, respectively, with $\hbar = 1$ for brevity. The resonator (H_r) acts as a quantum harmonic oscillator (see Equation 7) with a frequency of ω_r . The $\hat{a}^\dagger$ ($\hat{a}$) are the raising (lowering) operators which add (remove) a photon to the resonator mode.

The qubit, described by H_q , acts as a two-level system, assuming that higher excited states are inaccessible. Its energy splitting is determined by its transition frequency ω_q , which corresponds to the energy difference between the $|0\rangle$ and $|1\rangle$ states. This energy is described by the $\hat{\sigma}_z$ operator, whose eigenvalues $\{-1, 1\}$ correspond to the two computational basis states.

The final term, H_{int} , describes the interaction between the qubit and the resonator. The coupling strength g determines how strongly the two systems interact. The $\hat{\sigma}_+$ ($\hat{\sigma}_-$) are the raising (lowering) operators for the qubit, meaning that the terms $\hat{\sigma}_+ \hat{a}$ and $\hat{\sigma}_- \hat{a}^\dagger$ represent an exchange of excitations between the qubit and the resonator.

In the regime where the frequency difference between the qubit and resonator is small ($\Delta = |\omega_q - \omega_r| \ll g$), the qubit and resonator swap excitations coherently, allowing them to exchange energy. This regime is not suitable for a QND measurement as the resonator can alter the state of the qubit.

4.5 Dispersive Readout

In the regime where the frequency difference between the qubit and resonator is large ($\Delta \gg g$), known as the dispersive regime, the qubit and resonator interact without directly exchanging energy. In the limit that g/Δ is small, the Jaynes-Cummings Hamiltonian reduces to the dispersive approximation given by

$$H_{\text{disp}} = (\omega_r + \chi \hat{\sigma}_z) \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \frac{\tilde{\omega}_q}{2} \hat{\sigma}_z \quad (12)$$

where $\chi = g^2/\Delta$ is known as the dispersive shift, which quantifies the shift in the resonator frequency, and $\tilde{\omega}_q = \omega_q + g^2/\Delta$ is the Lamb shift, which quantifies the shift in the qubit frequency. Importantly, the effective resonator frequency ($\omega_r + \chi \hat{\sigma}_z$) is equal to $(\omega_r + \chi)$ if the qubit is in $|1\rangle$, and $(\omega_r - \chi)$ if the qubit is in $|0\rangle$. This means that the qubit state is encoded in the frequency shift of the resonator, which can be probed through a classical measurement.

These state-dependent changes to the resonator are probed using microwave spectroscopy, in which a transmission line is capacitively coupled to the resonator (see Figure 4.3). A short microwave pulse is applied at the input port, and the transmitted or reflected signal containing information about the qubit state is measured.

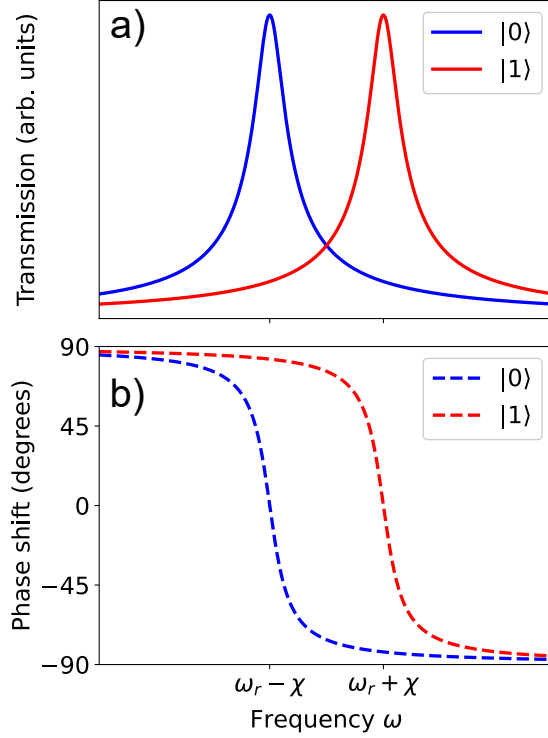


Figure 4.2: (top) Transmission of resonator and (bottom) phase shift of transmitted pulse as a function of the incident pulse frequency and qubit state. The parameters for the plots are adapted from [57].

Figure 4.2a shows an example transmission spectrum as a function of the probe frequency ω . The resonator's response exhibits Lorentzian peaks centered at $\omega_r - \chi$ and $\omega_r + \chi$, corresponding to the qubit being in the $|0\rangle$ and $|1\rangle$ state, respectively. This response is analogous to a classical damped driven harmonic oscillator, such as a mass-spring system with two different spring constants.

Figure 4.2b shows the phase shift of the transmitted signal as a function of the probe frequency ω_{RO} . The phase undergoes a transition of 180° near the resonance frequencies $\omega_r - \chi$ and $\omega_r + \chi$, corresponding to the $|0\rangle$ and $|1\rangle$ states, respectively. Below resonance, the transmitted signal lags behind the input signal resulting in a positive phase shift, and above resonance, the transmitted signal leads the input signal, resulting in a negative phase shift.

However, to determine the qubit state in a single shot, the probe must be at a fixed frequency. One approach is to set the probe at one of the shifted resonant frequencies $\omega_r - \chi$ or $\omega_r + \chi$ and measure the transmitted amplitude. For example, if $\omega_{RO} = \omega_r + \chi$, the transmitted amplitude will be high for $|1\rangle$ and low for $|0\rangle$, allowing the two qubit states to be discriminated. Another approach is to set the probe frequency to the bare resonator frequency ω_r , and then measure the phase shift of the transmitted signal, which is the approach used in this work. This process is depicted in Figure 4.3, including circuit diagrams of the qubit, resonator, and the input and output ports of the transmission line.

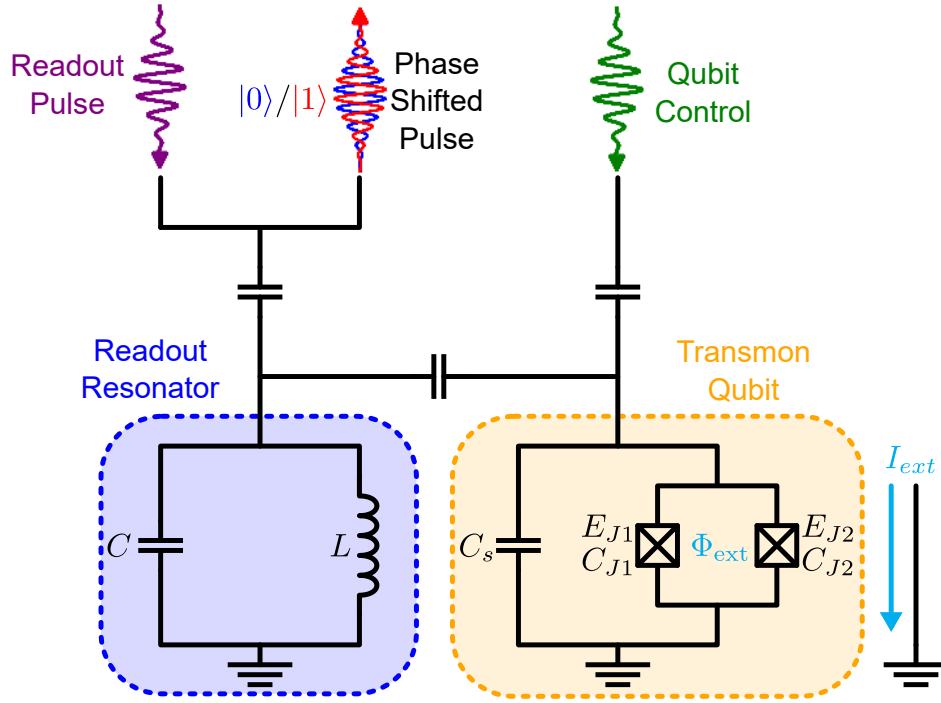


Figure 4.3: Circuit Diagram of the readout of a Transmon qubit coupled to a resonator, measured in transmission.

4.6 The Readout Chain

Before moving forward with demodulation and signal processing techniques, it is important to mention some additional details related to readout that are present in most experiments. See [18] for a comprehensive discussion on the following details.

The readout pulse is typically generated at room temperature using an arbitrary waveform generator (AWG) and undergoes frequency upconversion before being sent to the resonator. As it travels through the input line, it passes through a series of attenuators at progressively lower temperatures, which help suppress thermal noise and regulate the pulse power.

After interacting with the resonator, the transmitted signal containing information about the qubit state is typically weak and requires amplification. This is achieved using one or more amplifiers, such as a Josephson Parametric Amplifier (JPA) or a High Electron Mobility Transistor (HEMT), which operate at cryogenic temperatures to provide high gain while minimizing added noise. To enable faster readout while suppressing qubit decay through the readout line, a Purcell filter is often used to enhance signal transmission and protect qubit coherence.

The amplified signal is then typically mixed with a local oscillator (LO) in a detection scheme, either homodyne or heterodyne, frequency downconverting the signal before digitized by an analog-to-digital converter (ADC) and processed using control hardware, such as a field-programmable gate array (FPGA). The specifics of homodyne detection are discussed in the next section.

4.7 Homodyne Measurement

After amplification, the state-dependent phase shift can be detected by mixing the transmitted signal with a local oscillator using an in-phase and quadrature (I-Q) mixer, pictured in Figure 4.4. This process separates the signal into two components: the in-phase (I) component, which is aligned with the local oscillator, and the quadrature (Q) component, which is 90 degrees out of phase.

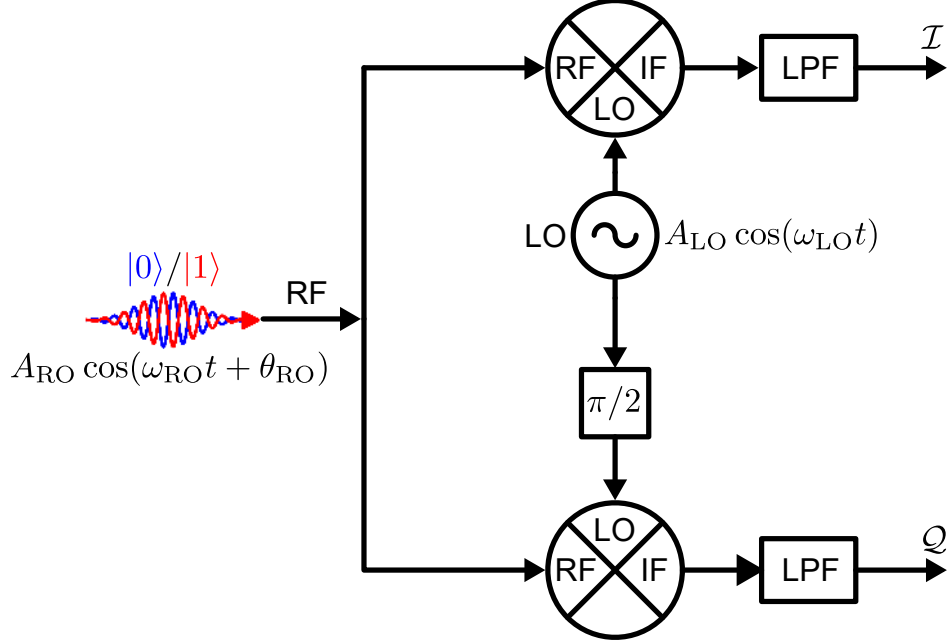


Figure 4.4: Analog I-Q Mixer.

In a homodyne measurement, the local oscillator has a frequency ω_{LO} equal to the carrier frequency ω_{RO} , causing $I(t)$ and $Q(t)$ to contain both DC components and oscillatory components at twice the carrier frequency. The signals are then averaged over an integer multiple of the local oscillator period, acting as a low-pass filter to isolate the DC components, and then sampled by an analog-to-digital converter (ADC). The resulting (I, Q) points form vectors in the I-Q plane, with amplitude $A_{RO} \propto \sqrt{I^2 + Q^2}$ and phase $\theta_{RO} = \arctan(Q/I)$, measured relative to the I-axis. The location of these vectors in the I-Q plane differs based on the qubit state, allowing for state discrimination, as depicted in Figure 4.5.

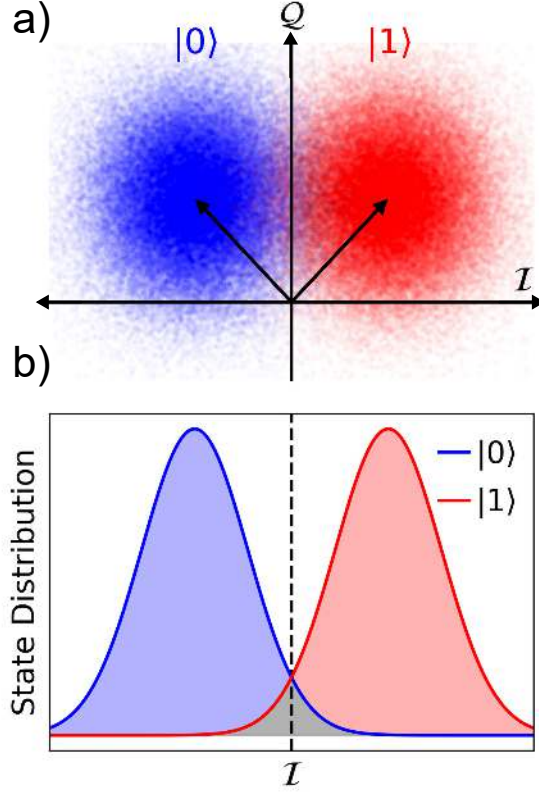


Figure 4.5: Example I-Q blobs. (a) Example readout measurements for 100K shots. (b) Distribution of states and decision boundary.

Over many repeated measurements, two Gaussian-like distributions emerge in the I-Q plane, corresponding to the two possible qubit states. A decision boundary can be drawn to separate these distributions and classify the state based on measurement outcomes. However, where the distributions overlap, misclassification errors occur (see gray region in Figure 4.5b), reducing the overall measurement fidelity. Exploring approaches to increase this fidelity, despite overlapping distributions, is a key focus of this thesis.

4.8 Frequency Multiplexed Readout

Up to this point, the readout of only a single qubit coupled to a resonator has been discussed, but quantum computers require readout of a large number of qubits. To save resources, the states of multiple qubits are typically measured simultaneously in a process known as frequency multiplexed readout, in which the resonators are coupled to a shared transmission line. Each resonator is designed with a unique resonance frequency, allowing a probe signal containing multiple frequency components to interact with all resonators simultaneously. The transmitted signal then carries qubit state information encoded in the phase and amplitude of each frequency component, which can be extracted using heterodyne detection.

Different from homodyne detection, the local oscillator frequency is different from the probe frequency, resulting in $I(t)$ and $Q(t)$ signals with frequency components at $\omega_{IF} = |\omega_{RO} - \omega_{LO}|$ and $\omega_{IF} = \omega_{RO} + \omega_{LO}$. The high frequency components can then be filtered out through time averaging. In multiplexed readout, the transmitted signal corresponding to each qubit can be isolated using different local oscillators.

One approach to reducing hardware resources is to downconvert the multiplexed signal with a single local oscillator, digitize it using an analog-to-digital converter (ADC), and then perform digital demodulation in the control hardware to extract individual qubit signals. Digital demodulation is achieved by multiplying the digitized signals with different numerically controlled oscillators (NCOs), reducing the need for multiple analog I-Q mixers at the cost of additional multipliers and computational resources in the control hardware, such as an FPGA. This trade-off, which impacts the computational complexity of qubit state discrimination, is discussed in detail in the next chapter.

While frequency multiplexed readout reduces resources, it also introduces crosstalk, where signals from different qubits unintentionally interfere due to shared components in the readout chain. This correlation between I-Q signals can complicate qubit state discrimination, as traditional methods often fail to account for crosstalk. As a result, more advanced models, including

machine learning-based approaches, may be necessary. Both traditional and machine learning-based discrimination methods are explored in detail in the next chapter.

4.9 Processing time-dependent I-Q signals

Until this point, the I-Q signal corresponding to a readout measurement have been treated as individual points in the I-Q plane, as shown in Figure 4.5a. In practice, these signals are continuously sampled throughout the readout duration using an analog to digital converter (ADC). These sampled signals trace out paths in the I-Q plane rather than occupying a single point. Both traditional and machine learning approaches utilize this temporal information to improve qubit state classification. Developing an efficient machine learning model to process this time-series data is a central focus of this thesis. A simplified overview of the readout process with time-series processing is depicted in Figure 4.6.

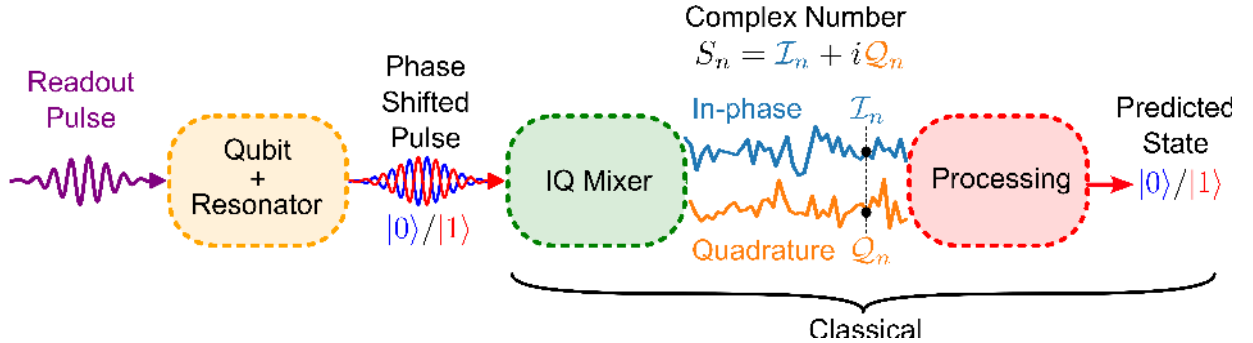


Figure 4.6: Simplified readout diagram for a transmon qubit dispersively coupled to a resonator.

4.10 Summary

This chapter provided an overview of superconducting qubit readout, including Transmon qubits, dispersive readout, the experimental readout chain, homodyne measurement, and frequency multiplexed readout. It also introduced the key challenge of processing time-series I-Q data for qubit state discrimination.

These details are essential for understanding the single and multi-qubit readout experiments described in the next chapter, as well as the traditional and machine learning approaches for state

discrimination presented in the next chapter. The key trade-off explored in the next chapter relates to the classification accuracy vs. the computational complexity of the qubit state discriminator.

Chapter 5

Superconducting Qubit Readout using Next-Generation Reservoir Computing

As previously discussed, superconducting quantum computers rely on fast, accurate, and scalable qubit state readout to realize their full potential. While traditional algorithms are computationally efficient, they cannot account for crosstalk and lack nonlinear processing, which reduces fidelity. Recent machine learning methods improve fidelity but come with significant computational overhead during training and evaluation, making it difficult to adapt to changing conditions, increasing latency and the risk of decoherence, and limiting scalability to larger systems. This chapter explores a novel approach to superconducting qubit state discrimination using a next-generation reservoir computer (NG-RC), which combines the computational efficiency of traditional methods with the high fidelity of machine learning techniques.

In this chapter, I provide background on the single-qubit and multi-qubit experimental setups, as well as traditional and machine learning approaches to qubit state discrimination. Then, I describe the design of the qubit state discriminator based on next-generation reservoir computing, as well as some useful performance metrics. Then, I discuss the results for both the single-qubit and multi-qubit datasets, focusing on fidelity, crosstalk reduction, and computational complexity. Last, I discuss the implications of these results.

5.1 Background

In this next section, I provide background on the single qubit and multi-qubit experimental setups, traditional discriminator approaches, and recent machine learning-based discriminators. I then describe my design based on next-generation reservoir computing. Following this, I present my results on classification fidelity and computational complexity and discuss their implications.

5.2 Single Qubit Experimental Setup

I apply the NG-RC algorithm to three single-qubit datasets analyzed by Khan *et al.* [23]. Each dataset consists of readout measurements from single qubit (labeled either A or B) dispersively coupled to a stripline resonator. Readout begins by generating a classical electrical pulse, which is upconverted to the resonator’s operating frequency and injected into the resonator. The reflected signal is amplified using a Josephson parametric amplifier (JPA) and frequency downconverted to extract $\mathcal{I}$ and $\mathcal{Q}$.

The first dataset uses qubit A, which has a short coherence time $T_1 = 9.5 \mu\text{s}$, and a relatively high $\chi/\kappa \approx 0.78$, where κ is the cavity loss rate, increasing its sensitivity to dephasing [23, 63]. It is more difficult to distinguish qubit A’s state because of its short coherence time. The readout pulse duration and amplitudes are varied for each set of M measurements.

The second dataset uses qubit B, which has the highest T_1 of $73 \mu\text{s}$ and a lower $\chi/\kappa \approx 0.20$. This data was collected with a fixed moderate readout pulse amplitude, and the phase between the JPA pump and readout pulse was varied after every set of M measurements.

The last dataset also uses qubit B but under different measurement conditions, which shortened T_1 to $60 \mu\text{s}$. Here, stronger readout pulses drive transitions to higher-energy states, creating a challenging three-state classification problem for which an optimal linear filter is not known [23].

5.2.1 Multi-Qubit Experimental Setup

I also use the NG-RC algorithm to classify the states of a five-qubit processor reported by Lienhard *et al.* [20]. Here, $7 \mu\text{s} < T_1 < 40 \mu\text{s}$, each qubit is dispersively coupled to its own coplanar waveguide resonator, and the resonators are inductively coupled to a shared Purcell-filtered bus.

The frequency multiplexed readout pulse has superposed baseband signals at intermediate frequencies (IFs). The composite signal is upconverted with a local oscillator to match the individual resonator frequencies and then injected into the signal bus. The transmitted signal containing the phase shifted tones is amplified and downconverted before being digitized to obtain $\mathcal{I}$ and $\mathcal{Q}$. The

signals corresponding to the individual qubits can be digitally demodulated using the IFs.

For a five-qubit system, there are 32 possible permutations of the basis states. To collect the dataset, the qubits are sequentially initialized in each permutation, and readout was performed 50,000 times for each configuration, with each readout lasting $2 \mu\text{s}$. For the results presented below, only the first $1 \mu\text{s}$ of the measurement is used to match Refs. [20, 24], and the ordering of the measurements are randomized, with 15,000 measurements per configuration reserved for training the algorithms and 35,000 for evaluating their performance.

5.2.2 Traditional approaches to qubit discrimination

The boxcar filter is one of the simplest and most widely used methods for qubit state discrimination, which sums the measured signal over a time window. The filtered signal is represented by a complex number in the $\mathcal{IQ}$ -plane as

$$\mathcal{S}_{\text{boxcar}} = \sum_n^N [S]_n = \sum_n^N (\mathcal{I}_n + i\mathcal{Q}_n), \quad (13)$$

$i = \sqrt{-1}$ is the imaginary unit. A discriminator is applied to the filtered value to determine the qubit state.

The matched filter extends the boxcar filter by introducing complex weights k_n as

$$\mathcal{S}_{\text{MF}} = \sum_n^N k_n [S]_n = \sum_n^N k_n (\mathcal{I}_n + i\mathcal{Q}_n), \quad (14)$$

which are chosen to maximize the SNR, which is given by

$$k_n = \frac{\langle [S_0]_n - [S_1]_n \rangle}{\text{var}([S_0]_n) + \text{var}([S_1]_n)} \quad (15)$$

for two-class discrimination. Here, S_0 and S_1 are the signals for the ground and excited states respectively, $\langle \dots \rangle$ denotes the mean value, and $\text{var}(\dots)$ denotes the variance. Equation 15 assumes Gaussian-distributed noise in the signal, which is not always accurate [23]. Like the boxcar filter,

a discriminator is applied to the filtered value to obtain the predicted qubit state.

The boxcar and matched filter require the individual qubit signals, which must be obtained through demodulation for a multi-qubit system with frequency multiplexing, requiring additional computations. Furthermore, the matched filter requires one multiplication for each n . However, these traditional approaches typically require substantially fewer operations than ML-based algorithms.

5.2.3 Machine learning approaches to qubit discrimination

Recently, Khan *et al.* [23] demonstrated that a simple ML-based algorithm, which they call the trainable temporal postprocessor (TPP), can outperform traditional approaches by avoiding assumptions about noise distribution and classifying an arbitrary number of states. Their approach finds a linear combination of $\mathcal{I}_n$ and $\mathcal{Q}_n$ at all times t_n in the readout signal. This combination maps to a 'one-hot' encoding of the qubit basis states, such as $[1, 0]$ for $|0\rangle$ and $[0, 1]$ for $|1\rangle$, where the index with the higher value represents the predicted qubit state.

The weights in the linear combination are determined using supervised learning (see Chapter 2.1), which is a type of ML that requires ground truth labels (*e.g.*, the true qubit states) for a subset of data described in Appendix 8.5. Their results are evaluated on the single qubit datasets described in Section 5.2.

The trained TPP algorithm has a complexity similar to a matched filter, making it computationally efficient to evaluate during deployment, and the optimal weights are found using linear regression, which is simpler than other supervised learning approaches. However, their model lacks nonlinear processing, which can limit its classification performance and ability to mitigate crosstalk in frequency multiplexed readout.

In contrast, other recent ML-based algorithms use feedforward neural networks (FNNs) [20, 24, 25, 64], described in Chapter 2.2.1. An FNN is an artificial neural network in which information flows from input to output. The network consists of an input layer, followed by a series of 'hidden layers' comprised of nodes or 'neurons' that perform a weighted sum followed by a

nonlinear activation function. The output layer encodes the qubit states, with the type of encoding affecting its size, as discussed later.

These FNNs are trained using supervised learning, which involves nonlinear optimization algorithms because the weights are embedded within nonlinear functions. This process is computationally demanding, often requiring hours of processing time on graphical processing units (GPUs). Additionally, these models frequently have millions of parameters, increasing the computational cost of optimization.

The main advantage of these FNN approaches is their ability to classify multi-qubit states with much higher accuracy than traditional methods, thanks to their use of nonlinear processing and ability to detect and mitigate crosstalk. However, they require a large number of multiplications and nonlinear operations, which demand significant hardware and computational resources. This makes them challenging to implement on modern control hardware like FPGAs, limiting their scalability to larger qubit systems. Additionally, the required computations introduce additional latency, increasing the risk of decoherence.

In one implementation by Lienhard *et al.* [20], the inputs to the FNN are the raw $\mathcal{I}_n$ and $\mathcal{Q}_n$ at all times t_n , eliminating the need for digital demodulation. Despite this simplification, evaluating the model consumes most of the computational resources on a high-end FPGA [24]. Moreover, one-hot encoding causes the output size to grow exponentially with the number of qubits, making it impractical for larger qubit systems.

Satvik *et al.* [24] addressed some of these challenges by demodulating the readout signal and then applying a matched filter to pre-process the data, reducing the number of inputs. While this lowers the complexity, it significantly reduces the classification fidelity. To improve fidelity, they introduced a relaxation-matched filter (RMF), which detects qubit relaxation from the excited to the ground state, but this requires manually identifying relaxation events in the training data.

Satvik *et al.* also use the MF and RMF to preprocess the inputs to a linear support vector machine (SVM), which is an ML algorithm that identifies an optimal hyperplane that best separates the data. The hyperplane is defined by the points closest to it, known as support vectors. Finding

them is costly for large datasets, and makes real-time training impractical. Despite the training cost, Gautam *et al.* [25] can evaluate a trained SVM readout algorithm operating on an FPGA in under 6 ns.

In addition, Gautam *et al.* [25] explored simplifying FNNs by reducing the model size and quantizing the model. Despite these improvements, they report a latency of 50 ns for their FPGA implementation, which is significantly longer than the SVM due to the large number of required operations that cannot be evaluated in parallel.

Mude *et al.* [65] build on Ref. [24] by introducing an excitation matched filter (EMF) alongside the MF and RMF, followed by a compact neural network. Their approach achieves higher fidelity with lower latency and resource usage than Refs. [20, 24] when implemented on an FPGA for a three-state classification task. However, the clustering algorithms used to identify relaxation and excitation events, along with the training procedure, are computationally intensive, making real-time training impractical.

In the next section, I present my approach, which combines the computational efficiency of matched filters with the nonlinear processing capabilities of FNNs, resulting in a fast algorithm that is scalable to large-qubit systems.

5.3 Next-Generation Reservoir Computer Design

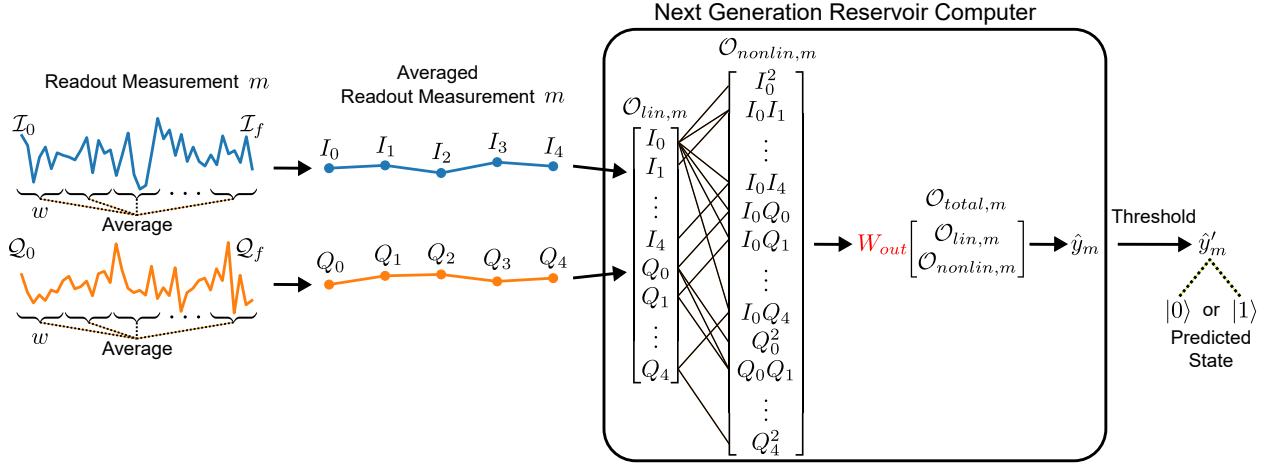


Figure 5.1: Superconducting qubit readout using the NG-RC algorithm followed by a discriminator.

The NG-RC, described in Chapter 2.4 and shown previously in Fig. 5.1, is an ML model well adapted for processing time-series data. The nonlinear functions can take on any form, but are often taken as monomials of the input variables at different times (see $\mathcal{O}_{nonlin,m}$ in Fig. 5.1). This enables the model to detect correlations between variables at different times [66], and maps the inputs to a higher-dimensional space to enhance its processing capacity. An advantage of this model is that the nonlinear features can be computed in parallel to reduce the latency, unlike in the neural networks that must evaluate each layer before continuing.

One downside of using monomials is that the number of features grows exponentially with the maximum polynomial degree. One method of reducing the number of monomials is to average $\mathcal{I}_n$ and $\mathcal{Q}_n$ over non-overlapping temporal windows of size w and computing the polynomial features from the averaged signals $I_{n'}$ and $Q_{n'}$. There is a trade-off between model fidelity and the size w as I discuss in Sec. 5.5. Another approach is to use nonlinear system identification techniques [34, 67], which selects the most important terms to retain. This is discussed in Appendix 8.4.

As shown in Fig. 5.1, the linear and nonlinear features are concatenated into the total feature vector $\mathcal{O}_{total,m}$. This vector is multiplied by the weights W_{out} found in the training phase, discussed

in Appendix 8.5, producing an output $\hat{y}_m$. For binary classification tasks, such as the first two single-qubit datasets, a threshold is applied to obtain the predicted state $\hat{y}'_m \in \{0, 1\}$. For the three-state classification problem in the third dataset, a one-hot encoding is used for $\hat{\mathbf{y}}_m$, and the predicted state $\hat{y}'_m$ is obtained by applying the *argmax* function, which assigns the highest probability to the largest output component.

For the multi-qubit algorithm, the readout signals are demodulated to extract $(\mathcal{I}_n, \mathcal{Q}_n)$ for each qubit. A binary mask sets these values to zero past a specific time for each qubit, where the signal no longer provides useful information about the qubit state. Next, the number of linear features is reduced by averaging over w . I create a feature vector that includes data from all qubits to detect and account for crosstalk. For multiplexed readout with more qubits, scalability can be achieved by using only data from qubit signals with nearby resonator frequencies, as they experience the most crosstalk.

I train five distinct NG-RC models, each tasked with predicting the state of a single qubit. Like the other binary state classifiers, the output is a single value $\hat{y}_m$, which is discriminated to yield a binary state (0 or 1) and a discriminator threshold optimized to maximize the fidelity. This avoids the exponential growth of the output size seen in Ref. [20].

5.4 Readout Measurement Metrics

I use several metrics to assess the performance of the NG-RC algorithm with other methods. For all datasets, I use the classification fidelity $\mathcal{F}$, which is the ratio of the number of correct classifications to the total number of classifications. This metric ranges from 0 to 1, with values closer to 1 indicating higher classification accuracy.

I define the infidelity reduction as the relative decrease in classification error when using a machine learning method instead of the matched filter:

$$\eta = \frac{(1 - \mathcal{F}_{MF}) - (1 - \mathcal{F}_{ML})}{1 - \mathcal{F}_{MF}}. \quad (16)$$

Here, $\mathcal{F}_{MF}$ and $\mathcal{F}_{ML}$ are the fidelities of the matched filter and the machine learning method, respectively. Positive (negative) values of η indicate improvement (worse performance) over the matched filter.

For multi-qubit classifications of N_q qubits, the overall bit pattern is important for the quantum computation. To specify the fidelity of the pattern, I also use the geometric mean classification fidelity

$$\mathcal{F}_{GM} = (\prod_{r=1}^{N_q} \mathcal{F}_r)^{1/N_q}, \quad (17)$$

where $\mathcal{F}_r$ is the individual qubit fidelity. As before, values closer to 1 indicate higher classification accuracy.

To quantify the extent to which the model corrects for crosstalk, I define the cross-fidelity

$$\mathcal{F}_{jk}^{CF} = \langle 1 - [P(1_j|0_k) + P(0_j|1_k)] \rangle, \quad (18)$$

where $P(1_j|0_k)$ is the conditional probability that the model predicts qubit j to be in the excited state given that qubit k was prepared in the ground state, and $P(0_j|1_k)$ is the conditional probability that the model predicts qubit j to be in the ground state, given that qubit k was prepared in the excited state. Cross-fidelity values closer to zero indicate greater crosstalk reduction.

5.5 Results

5.5.1 Single Qubit

The single-qubit fidelity as a function of the number of multiplications in model evaluation is shown in Fig. 5.2a-c. Each panel shows fidelity curves corresponding to the measurement powers (a,c) or JPA phase (b) that yield the highest classification fidelity achieved across all models and measurement conditions. The goal is to maximize fidelity while minimizing the number of multiplications, which serves as a key indicator of model complexity. The curves with multiple points correspond to the different window sizes w tested, with larger w corresponding to fewer multipli-

cations. The linear NG-RC with the most multiplications corresponds to no time averaging.

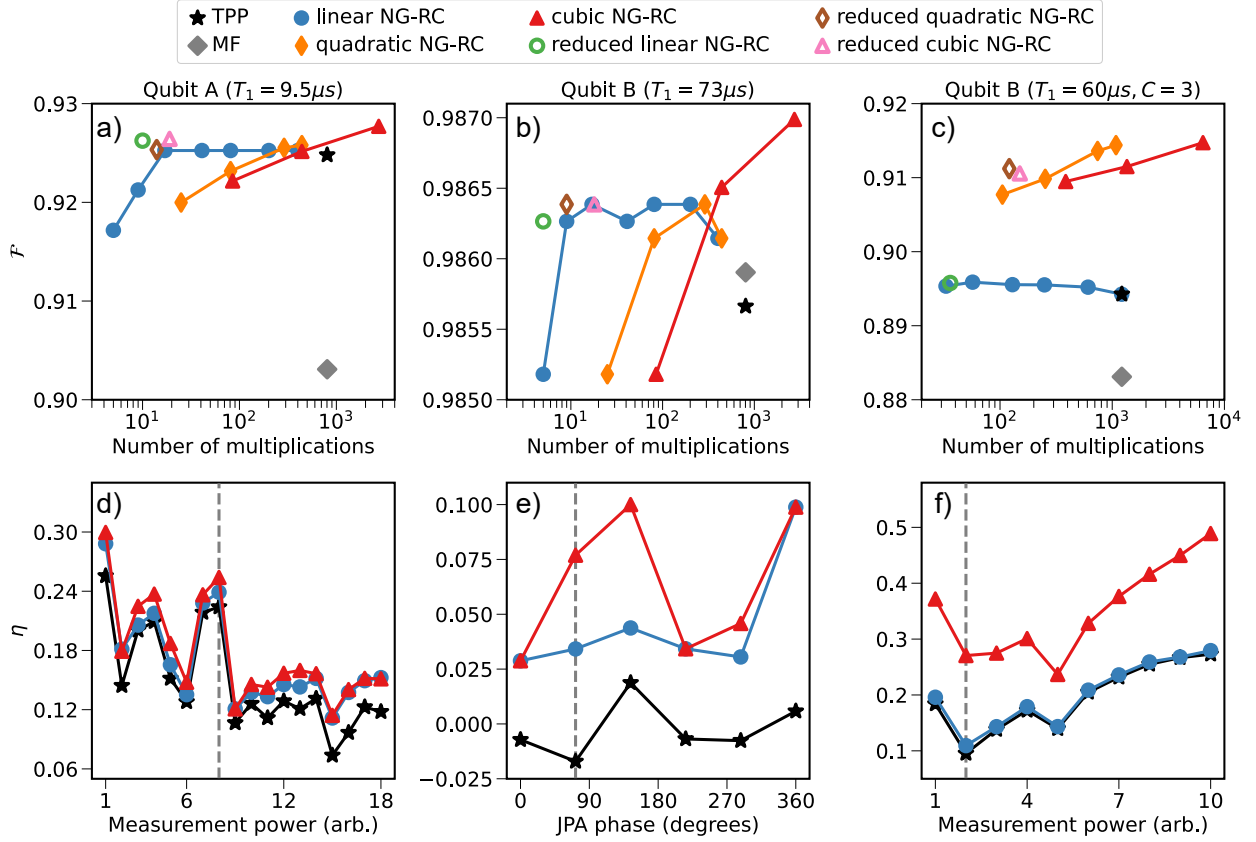


Figure 5.2: Classification fidelity and infidelity reduction for single qubit datasets. The matched filter (MF) and temporal postprocessor (TPP) results are taken from Ref. [23]; all other data points are from this work. (a)-(c) Classification fidelity as a function of the number of features in the model for each of the three datasets, where the measurement conditions correspond to the dashed gray lines in (d)-(f), which show the infidelity reduction compared to the MF as a function of the measurement conditions. In the bottom row, the black stars represent the TPP, the blue circles denote the best linear NG-RC, and the red triangles indicate the best nonlinear NG-RC.

In (a-b), the linear NG-RC without averaging requires fewer multiplications than the TPP due to its single-output encoding, whereas TPP uses one-hot encoding. For dataset (c), both methods use one-hot encoding, resulting in equal complexity and accuracy. For the JPA phase shown in (b), the TPP has a lower fidelity than the MF, but this dataset has the lowest variation in fidelity.

Notably, increasing averaging for the linear NG-RC (a-c) does not result in strictly declining fidelity. Instead, the fidelity reaches a peak before eventually declining. Moreover, performing system identification to reduce the number of features provides the highest fidelity of the linear

models for both (a) and (c), requiring only a few multiplications. This robustness to averaging indicates that a linear NG-RC can achieve higher fidelity than the TPP and MF with a significantly lower complexity.

We see in (a-c) that the quadratic and cubic NG-RCs achieve a greater maximum fidelity than any of the linear models. While this increase is less pronounced for (a) and (b), the nonlinear features provide a significant jump in fidelity for the three-state classification task in (c). Importantly, this increase in fidelity occurs at a similar number of multiplications as in the linear models. The reduced quadratic and cubic features have similar fidelity to the best linear NG-RC in (a-b), but exhibit noticeable reductions in complexity compared to the full nonlinear models for (c). These results indicate that nonlinear features provide a significant increase in fidelity for the most challenging single-qubit classification task without compromising computational efficiency. The nonlinear features account for temporal correlations in the data, which I hypothesize is responsible for the increased fidelity.

Figure 5.2d-f shows the infidelity reduction (16) relative to the MF for the TPP, the best linear NG-RC, and the best nonlinear NG-RC for all measurement powers (d,f) and JPA phases (e). The vertical dashed lines indicate the measurements that correspond to the example fidelity curves shown in Fig. 5.2a-c. Here, a larger infidelity reduction indicates better performance. Notably, the infidelity reductions are high for all models in (d,f), but remain small in (e), likely because the Gaussian noise assumption of the matched filter holds in that case.

In (d-f), we see that the linear NG-RC has a higher infidelity reduction than the TPP for all measurement conditions, with the TPP performing worse than the matched filter for some JPA phases. The best nonlinear NG-RC always matches or outperforms the best linear NG-RC, with the highest infidelity reduction of nearly 50% occurring for the three-state task in (f). These results highlight that the NG-RC provides fidelity gains compared to the TPP and matched filter across a wide range of measurement conditions.

5.5.2 Multi-Qubit

The fidelity as a function of the number of multiplications is shown in Fig. 5.3a-e. Like the single qubit results, a curve with multiple points indicates that multiple window sizes are evaluated, with larger w corresponding to fewer multiplications. The number of multiplications indicates the value needed to predict all five qubit states. The goal is to maximize the fidelity while keeping the number of multiplications low.

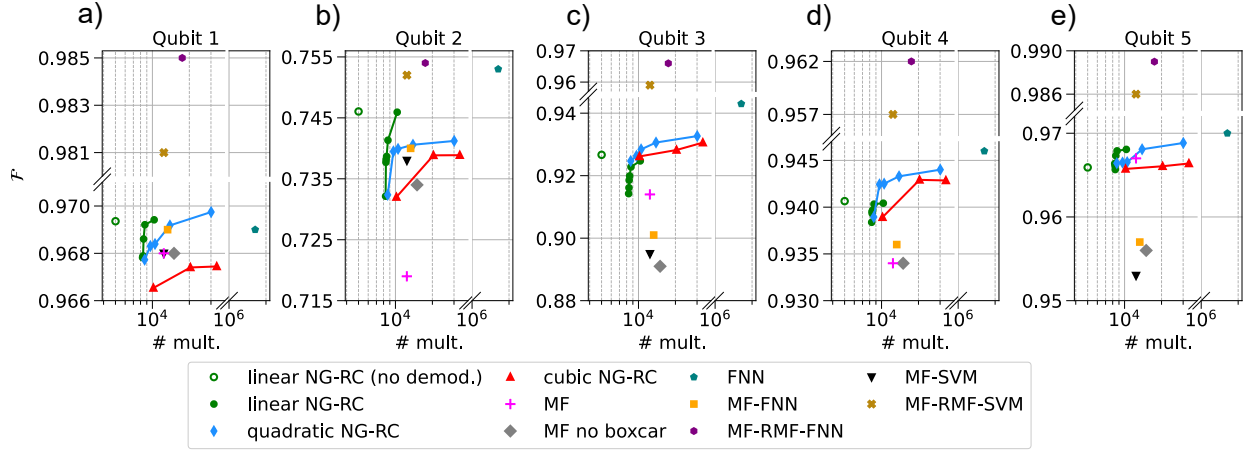


Figure 5.3: Classification fidelity for a 5-qubit chip. Panels (a)–(e) show the classification fidelity for qubits 1 through 5, respectively. The MF and FNN results are taken from Ref. [20], and MF (no boxcar), MF-FNN, MF-RMF-FNN, MF-SVM, and MF-RMF-SVM are taken from Ref. [24]. All other results are from this work.

We see that the linear NG-RC, without digital demodulation or window averaging, requires the fewest multiplications, yet achieves higher fidelity than the matched filter, an FNN, and an SVM. This highlights that a simple NG-RC model can outperform both traditional methods and some ML approaches.

For the linear NG-RC with demodulation, we see that increasing w for averaging strictly decreases the fidelity. Interestingly, it has a higher maximum fidelity than the linear NG-RC without demodulation for qubit 5.

For the quadratic and cubic NG-RCs, we see that they generally achieve a higher maximum fidelity than the linear NG-RC except for qubit 2, which has significantly lower fidelity than the others. The quadratic NG-RC outperforms the cubic NG-RC across all qubits, as the cubic model

uses larger window sizes for averaging. This keeps the number of multiplications manageable but results in reduced fidelity.

Importantly, the NG-RC outperforms the baseline FNN for qubit 1, with only slightly lower fidelity for the other qubits, while requiring orders-of-magnitude fewer operations. While the MF-RMF-SVM and MF-RMF-FNN algorithms achieve higher fidelity than the baseline FNN with reduced complexity, I attribute this increase in performance to the RMF detecting relaxation events.

These results indicate that the NG-RC can outperform the traditional and some ML approaches while requiring fewer multiplications. In addition, I conclude that including nonlinear terms in the model can result in an increase in fidelity without substantial increases in complexity.

The maximum fidelity, as well as the geometric mean fidelity (Equation 17) achieved by each approach is summarized in Table 3. Notably, all NG-RC variations perform better than the traditional approaches, as well as the MF-SVM and MF-NN, which are both machine learning based approaches with higher computational complexity.

Design	Qubit 1	Qubit 2	Qubit 3	Qubit 4	Qubit 5	F_{5Q}
MF [24]	0.968	0.734	0.891	0.934	0.956	0.892
MF + boxcar [20]	0.968	0.719	0.914	0.934	0.967	0.895
MF-SVM [24]	0.968	0.738	0.895	0.928	0.953	0.892
MF-NN [24]	0.969	0.740	0.901	0.936	0.957	0.896
Cubic NG-RC $w = 200$	0.967	0.739	0.931	0.943	0.966	0.905
Linear NG-RC (no demod.)	0.969	0.746	0.927	0.941	0.966	0.906
Linear NG-RC $w = 10$	0.969	0.746	0.925	0.940	0.968	0.906
Quadratic NG-RC $w = 50$	0.970	0.741	0.933	0.944	0.969	0.907
NN [20]	0.969	0.753	0.943	0.946	0.970	0.912
MF-RMF-SVM [24]	0.981	0.752	0.959	0.957	0.986	0.923
MF-RMF-NN [24]	0.985	0.754	0.966	0.962	0.989	0.927

Table 3: Individual qubit readout fidelities and geometric mean fidelity (F_{5Q}) for different designs.

The cross-fidelity is shown in Table 4 for different hamming distances, defined as the separation between qubit indices, with lower values indicating better crosstalk mitigation. We see that the NG-RCs have better crosstalk mitigation than the MF and two ML methods, with the quadratic

NG-RC achieving similar performance to the neural networks in Refs. [20, 24]. I attribute this to the inclusion of cross-terms between individual qubit signals, which help detect crosstalk.

Design	$\langle F_{k=j\pm 1}^{CF} \rangle$	$\langle F_{k=j\pm 2}^{CF} \rangle$	$\langle F_{k=j\pm 3}^{CF} \rangle$	$\langle F_{k=j\pm 4}^{CF} \rangle$	$\langle F^{CF} \rangle_{\Delta k}$
NN [20]	0.002	0.005	0.002	0.0003	0.0023
MF-RMF-NN [24]	0.0031	0.0062	0.0008	0.0005	0.0027
Quadratic NG-RC $w = 50$	0.0037	0.0049	0.0020	0.0009	0.0029
Linear NG-RC (no demod.)	0.0036	0.0066	0.0024	0.0006	0.0033
Linear NG-RC $w = 10$	0.0048	0.0095	0.0013	0.0005	0.0040
MF-NN [24]	0.0071	0.011	0.003	0.0003	0.0054
MF-RMF-SVM [24]	0.011	0.0077	0.0024	0.0006	0.0054
MF [24]	0.0108	0.015	0.0021	0.0008	0.0072

Table 4: Mean absolute values of the cross-fidelity at different hamming distances, with $\langle |F^{CF}| \rangle_{\Delta k}$ indicating the average over all qubit index separations $\Delta k = |k - k'|$.

The computational complexity of the different models is shown in Table 5. The goal is to minimize the number of trainable parameters, multiplications, and nonlinear activation functions while maximizing the geometric mean fidelity. For models that have a matched filter or relaxation matched filter as input, the parameters needed to learn the matched filter are counted as well. This table does not include the computations required to train the model. The NG-RC or the traditional algorithms are much simpler to train than an FNN or SVM because they do not require nonlinear optimization. This suggests that the NG-RC training can be done directly on an FPGA in near real time.

Design	# Parameters	# Multiplications	# Activation Functions	F_{5Q}
MF [24]	5×10^3	1.5×10^4	0	0.892
MF + boxcar [20]	4.13×10^3	1.24×10^4	0	0.892
MF-SVM [24]	4185	1.24×10^4	0	0.892
MF-NN [24]	5082	1.33×10^4	62	0.896
Cubic NG-RC $w = 200$	1.83×10^4	3.01×10^4	0	0.905
Linear NG-RC (no demod.)	5000	5000	0	0.906
Linear NG-RC $w = 10$	55	8.27×10^3	0	0.906
Quadratic NG-RC $w = 50$	1.83×10^4	3.01×10^4	0	0.907
NN [24]	1.63×10^6	1.63×10^6	1782	0.912
MF-RMF-SVM [24]	8365	1.24×10^4	0	0.923
MF-RMF-NN [24]	9262	1.75×10^4	62	0.927

Table 5: Computational complexity of discriminators.

We see that the model that requires the fewest number of parameters is the linear NG-RC with an averaging window size of 10 time steps. This is orders of magnitude fewer than the other approaches, despite having a higher accuracy than traditional methods and those with MF pre-processing. We also see that the number of parameters for the quadratic and cubic NG-RC with the highest fidelity are large, but still fewer than the full NN [20].

We see that the full NN [20] requires a large number of nonlinear activation functions, which greatly increases the resources required to implement the model. While the other neural network approaches require fewer nonlinear function evaluations, they are still costly to implement. Notably, both the traditional and NG-RC approaches require no costly nonlinear activation functions.

5.6 Discussion

I show that a NG-RC can perform qubit state classification with higher fidelity than traditional methods and other ML approaches while requiring fewer operations. Additionally, I show that nonlinear terms can be included to increase fidelity further without significantly increasing the complexity of the model. Last, I show that system identification techniques can be used to further reduce model complexity.

Based on previous work implementing an SVM [25], which has a similar evaluation complexity to the NG-RC, I expect that my approach can be evaluated on control hardware such as an FPGA in a few nanoseconds. Furthermore, the batch training process for my approach is simple enough to be implemented in near real time to account for drifts in the system. Future work will add information about relaxation events to the feature vector, which I expect will increase the fidelity.

Chapter 6

Neutral Atom Readout

An emerging platform for quantum computing uses an array of trapped neutral atoms [26–29], where laser-cooled atoms are loaded into far-off-resonance traps. Qubits are typically encoded in two hyperfine ground states, manipulated by microwave or optical fields, and measured via resonance fluorescence.

This platform offers several advantages, such as naturally identical qubits, avoiding manufacturing variability present in superconducting qubit platforms. It is also highly scalable, as large numbers of atoms can be trapped, individually controlled, and dynamically rearranged using optical tweezers. In addition, neutral atom qubits exhibit long coherence times and can be excited to Rydberg states (principal quantum number $n \gg 1$), which interact through dipole-dipole interactions, enabling multi-qubit gates and the execution of quantum algorithms.

While neutral atom systems show great promise, accurate qubit state measurement remains a key challenge due to experimental imperfections, crosstalk, and the lack of scalable machine learning methods for high-accuracy state classification. These are all key problems that I address in the next chapter.

This chapter provides the necessary background for the results presented in the next chapter. This includes how the qubit states are encoded, how the atoms are cooled and trapped, and how their states are measured. For a more comprehensive discussion on neutral atom quantum computing, see Refs. [26, 27].

6.1 Qubit state encoding

In neutral atom quantum computing, the heavy alkalis Rubidium (Rb) and Cesium (Cs) are often chosen for their suitability for laser cooling, their large ground-state hyperfine splittings that allow for stable qubit states [26], and their optical transitions accessible with well-established laser tech-

nology. The results in the next chapter utilize Cs atoms, so the following discussion focuses on this species.

Hyperfine splitting occurs due to the interaction between the nuclear spin I and the total electronic angular momentum J , resulting in energy levels characterized by total angular momentum F , where $F = |I - J|, |I - J| + 1, \dots, I + J$. For Cs-133 with $I = 7/2$ and $J = 1/2$, this gives $F = 4$ (parallel spin) and $F = 3$ (anti-parallel spin).

The qubit states are encoded in the two hyperfine ground states, with $|0\rangle \equiv |F = 4, m_F = 0\rangle$ and $|1\rangle \equiv |F = 3, m_F = 0\rangle$, where m_F is the magnetic quantum number. It is useful to refer to these as the bright ($|B\rangle$) and dark ($|D\rangle$) states, respectively, for the readout discussion in the next section. Since they are both in the electronic ground state, they are not subject to spontaneous emission. As a result, the longitudinal relaxation time T_1 is limited only by collisions with stray gas particles in the vacuum or interactions with electromagnetic fields. Recent experiments have demonstrated transverse coherence times T_2 as long as 12.6 seconds [29].

6.2 Qubit Register Preparation

To prepare neutral atoms in an array, a dilute vapor is first produced inside an ultra-high vacuum chamber at room temperature. The atoms are then laser-cooled and trapped in a 3D magneto-optical trap, confining them to a $\sim 1 \text{ mm}^3$ volume [27, 68]. To isolate the individual atoms, a second laser system is focused using high numerical aperture lenses to create an array of optical tweezers, each capable of confining a single atom to a volume of a few μm^3 .

Initially, only about 50% of the traps contain an atom due to the probabilistic loading process. The locations of the occupied traps are identified using fluorescence imaging, as discussed in the next section. Once the occupied sites are known, atoms can be rearranged by dynamically moving the tweezers using devices such as acousto-optic deflectors (AODs) to create a localized, fully loaded array. Finally, the atoms are initialized in the desired computational basis states using a combination of optical pumping and single-qubit gates, allowing quantum processing to occur.

6.3 Readout

As mentioned before, qubit states are measured using resonance fluorescence, in which the atoms in $|B\rangle$ are illuminated with resonant light that drives cycling transition to an excited state $|e_B\rangle$, causing them to scatter photons [69]. The scattered photons are then detected by an electron-multiplying charge-coupled-device (EMCCD), producing an array of pixel values I_{ij} . Atoms in $|B\rangle$ appear as bright spots in the image, while atoms in $|D\rangle$ remain dark. These values are then processed to determine the qubit states, as discussed in the next section.

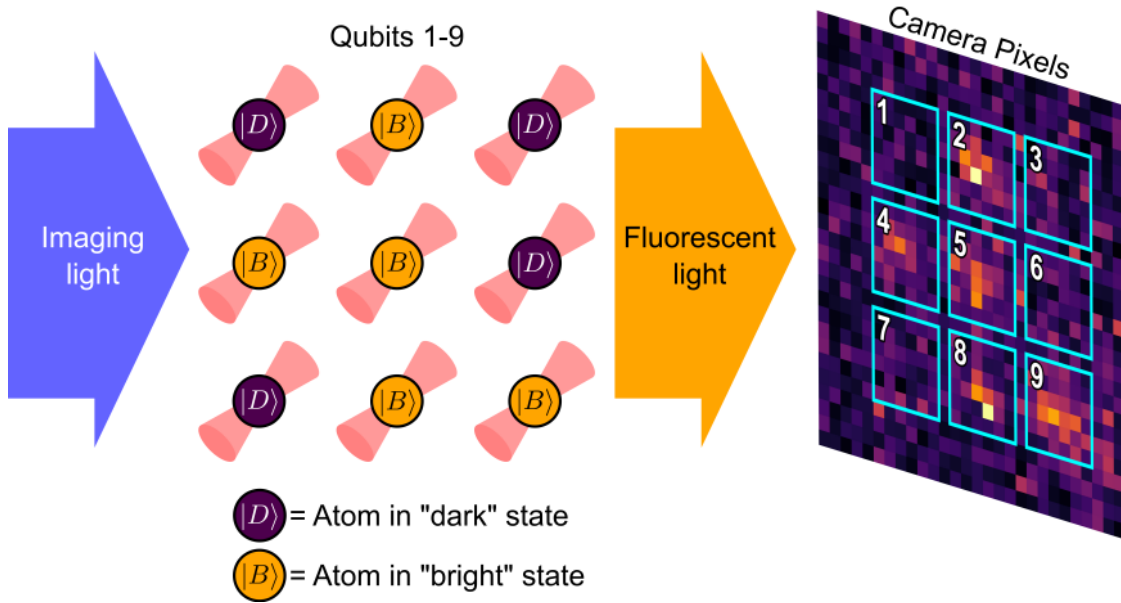


Figure 6.1: Diagram of readout process for an array of neutral atom qubits. The numbers inside the pixel boundaries represent qubits numbered 1-9.

However, if the cycling transition is not closed, atoms can undergo a Raman transition to D , causing them to stop fluorescing, which results in an incorrect state classification. Methods to minimize these transitions are discussed in Ref. [69]. This approach enables non-destructive measurement, as the qubits remain in their measured states and are reusable for further computation, but this comes at a decreased measurement fidelity.

To maximize the measurement fidelity, a destructive measurement approach is used, in which the atoms in $|B\rangle$ are ejected from their traps, and the remaining atoms in the $|D\rangle$ are optically

pumped into $|B\rangle$ for fluorescence detection. This simplifies readout from detecting the state-dependent fluorescence to detecting whether or not an atom remains in a particular trap. However, the ejected qubits cannot be reused, so this method is typically reserved for the end of a quantum computation.

One essential challenge in neutral atom qubit readout is crosstalk, where fluorescence from one atom impinges on pixels assigned to a neighboring atom. This can occur when the separation between atoms is small ($\lesssim 5 \mu\text{m}$), or due to imperfections in the optical setup, such as misalignment or aberrations. This introduces two problems: first, additional light from neighboring qubits makes it less likely to correctly classify a qubit as $|D\rangle$; second, reduced fluorescence from the qubit itself makes it more difficult to classify it as $|B\rangle$.

Another essential challenge is that the pixel values measured using the EMCCD are inherently noisy [69], which necessitates longer measurement times to collect enough fluorescence for confident state discrimination, typically tens of milliseconds. This is significantly longer than the time required to implement quantum gates, which take only a few microseconds [30]. A long measurement time can introduce additional errors, such as decoherence (T_2) or heating that can lead to atom loss.

The last challenge, which is perhaps the most crucial to the success of future neutral atom quantum computers, is designing an approach that can efficiently process the large array of pixel values I_{ij} from a readout measurement to classify the states of many qubits with high fidelity. Current machine learning approaches can achieve high fidelity on small neutral atom arrays [30], but are not scalable to larger qubit systems due to increasing computational demands for training and hardware constraints during evaluation.

6.4 Summary

This chapter provided background on how states are encoded in neutral atom qubits, how these atoms are cooled and trapped, and how they are rearranged to achieve higher occupancies. In addition, I discussed the readout process in detail for both destructive and non-destructive approaches,

as well as the core challenges such as crosstalk, noisy measurements, and scalability.

These details are essential to understand the contributions in the next chapter. For example, the Cs atoms are loaded in a 3×3 array, each with a randomly assigned state $|B\rangle$ or $|D\rangle$, and destructive readout is used to maximize measurement fidelity. In addition, these atoms have a spacing of $5\ \mu\text{m}$, resulting in crosstalk, and the classification fidelity as a function of the measurement time is explored. Lastly, I present a machine learning model that addresses these core issues by mitigating the effects of crosstalk, reducing the required measurement time, and scales efficiently to larger qubit systems. My approach, along with traditional and machine learning methods for readout, is discussed and compared in the next chapter.

Chapter 7

Neutral Atom Readout using Next-Generation Reservoir Computing

As previously mentioned, fast, efficient, and scalable qubit state discrimination is key to the success of quantum computers. For neutral atom systems, traditional discriminator approaches can be evaluated quickly but cannot account for crosstalk from neighboring qubits or experimental imperfections such as optical aberrations or noise. Recent machine learning approaches based on convolutional neural networks (CNNs) improve classification fidelity but are computationally expensive to train and evaluate. This complexity also scales poorly with increasing qubit count, making these approaches infeasible for the larger neutral atom computers of the future. This chapter explores a novel approach to neutral atom qubit discrimination based on next-generation reservoir computing (NG-RC) that is efficient to train and evaluate, and scalable for larger qubit systems.

In this chapter, I start by discussing background on the experimental setup, as well as the traditional approaches for qubit state discrimination. I then describe some useful performance metrics, followed by an overview of recent machine learning approaches. After that, I present two discriminator models based on NG-RC, one of which is specifically designed to mitigate crosstalk. I discuss the results, focusing on fidelity as a function of measurement time, crosstalk reduction, computational complexity, and how the weights offer physical interpretability to diagnose problems in the experimental setup. Finally, I conclude with a discussion of the implications of these results.

7.1 Background

In this section, I discuss the necessary background, including the experimental setup, traditional and machine learning discriminator designs, and useful metrics.

7.1.1 Experimental setup

My analysis is performed on the dataset measured and analyzed by Phuttitarn *et al.* [30]. Briefly, they load single cesium atoms in each of 9 far-off-resonance optical traps in a square 3×3 array. To read out the qubit state, all atoms in the array in one qubit state are pushed out of their trap using a large optical beam. Then, resonant laser light is applied to the remaining atoms, which fluoresce. Sites that are ‘dark’ indicate atoms that were in the pushed-out state, and ‘bright’ sites indicate atoms in the other state.

A high-numerical aperture lens images the fluorescence emitted by the trapped atoms onto an electron-multiplying charge-coupled device digital camera. They intentionally attenuate the light traveling along this path to simulate a noisy readout with a shorter exposure readout period and hence there is a reduced intensity for qubit readout. Here, I focus on the setup where the array spacing is $5 \mu\text{m}$, which is close enough that fluorescent light from one atom overlaps with its neighbors, causing crosstalk.

They also measure the fluorescent light from the opposite side of the trap without purposeful attenuation, which gives a higher signal-to-noise ratio and hence a higher fidelity readout for a given measurement time. This high signal-to-noise measurement path is used to determine the measurement fidelity for the first path and for training the machine-learning algorithms. See Ref. [30] for the details of the experimental setup.

7.1.2 Traditional qubit readout

The traditional approach for estimating the qubit state involves processing the pixel intensity data I_{ij} , which depends on the length of time the resonant optical fields illuminate the atoms. Longer readout times achieve higher state fidelity (defined below), but this slows down possible error correction protocols and the experimental cycle time.

The simplest approach is to find

$$S_{\text{square}} = \sum_{ij}^N I_{ij}, \quad (19)$$

where the indices are taken from a square surrounding each atom with N pixels. The central atom location can be determined by averaging many images of the array when the atoms are in different qubit states. The probability density of the sum $\mathcal{P}(S_{\text{square}})$ for the dark and bright states are each expected to be approximately Gaussian, and the optimum threshold for the state estimate is the value of S_{square} where the distributions intersect. For short readout times, the signal-to-noise of the images are low and there is substantial overlap in the distributions, which causes state misidentification.

A lower error algorithm assumes that the intensity profile from each atom has a Gaussian shape so that the sum takes the form

$$S_{\text{Gaussian}} = \sum_{ij}^N W_{ij} I_{ij}. \quad (20)$$

Here, W_{ij} are discrete values of a two-dimensional Gaussian function centered at the location of each atom, where the standard deviation σ is determined by fitting a circular Gaussian to the averaged images. In Ref. [30], these weights are defined across the entire image but remain small far from the center, where they can often be set to zero to reduce computational cost.

The key advantage of these approaches is that they do not require knowing the true qubit states. These algorithms are called ‘unsupervised’ in the machine learning field because they do not need ‘ground truth’ data for training (optimizing) the algorithm. However, because the filters are local to each qubit, they struggle to account for crosstalk.

7.1.3 Readout measurement metrics

One key metric to quantify the qubit state readout performance is the classification fidelity given by

$$\mathcal{F} = 1 - \frac{P(B_p|D) + P(D_p|B)}{2}, \quad (21)$$

where $P(B_p|D)$ is the probability that the model predicts the qubit to be in a bright state given that it was actually dark (false-bright), and $P(D_p|B)$ is the probability that the model predicted the qubit to be in a dark state given that it was actually bright (false dark).

Next, I define the cross-fidelity [70], which measures the predicted correlations between neighboring qubits, giving insight into the model's susceptibility to crosstalk. The cross-fidelity is given by

$$\mathcal{F}_{kl}^{\text{CF}} = 1 - \langle P(D_k|B_l) + P(B_k|D_l) \rangle, \quad (22)$$

where $P(D_k|B_l)$ is the probability that the model predicted the qubit at site k as dark given that the model predicted bright at site l , and $P(B_k|D_l)$ is the probability that the model predicted the qubit at site k as bright given that the model predicted dark at site l .

Lastly, I define the infidelity reduction, which represents the percent decrease in incorrect state predictions relative to the traditional Gaussian filter approach, as

$$\eta = \frac{(1 - \mathcal{F}_\sigma) - (1 - \mathcal{F}_{\text{MF}})}{1 - \mathcal{F}_\sigma} \quad (23)$$

where $\mathcal{F}_\sigma$ is the fidelity given by the traditional Gaussian filter approach, and $\mathcal{F}_{\text{MF}}$ is the fidelity given by the matched filter approach discussed in the next section.

7.2 Enhancing qubit readout using classical machine learning

Recently, Phuttitarn *et al.* [30] demonstrated that a CNN can mitigate crosstalk and increase measurement fidelity. This is achieved by learning nonlinear filters (their so-called CNN-Site model) or by processing a larger pixel area to account for neighboring qubits (CNN-Array model).

A CNN is a feed-forward artificial neural network (see Chapter 2.2.2), where information flows in a unidirectional manner from input to output [35]. It has many layers, each containing many ‘neurons,’ where a neuron performs a nonlinear transformation of weighted and summed input information. Common nonlinear transforms include sigmoids or rectifying piecewise-linear functions. Small filters are applied to the input image data to form a convolution between the image

and the filter; using several filters focuses model attention on different image features. The output of the network is mapped to numerical ‘classes’ representing the problem solution: identifying the dark (mapped to 0) and bright (mapped to 1) qubit states.

This approach typically relies on supervised learning, which requires that the true qubit labels (dark or bright) are known to train the model. In this work and Ref. [30], the true labels are determined by applying a Gaussian filter to the unattenuated path, but the machine learning model is trained to predict qubit states using only images from the attenuated path.

Each dataset that corresponds to a different measurement time is split into training, validation, and testing images: the training images are used for training or optimizing the neural network weights, the validation images are used to monitor the performance during training, and the testing set is used to evaluate the model’s performance on unseen data.

Nonlinear optimization algorithms are required for training the weights because they appear within nonlinear functions. Nonlinear optimization is computer intensive, often requiring hours of computing time using graphical processor units. Furthermore, the models tend to have many weights: their CNN-Array model requires tens of millions of parameters and 30 passes through the dataset during optimization.

Here, I develop two supervised machine learning algorithms that realize matched filters. My approach dramatically reduces the computational complexity of the model, both in the training and evaluation phases. They are linear models similar to the Gaussian filter (Eq. 20), but the weights W_{ij} are learned using regularized linear least-square regression. This machine learning model is faster to train because it only requires linear convex optimization with a guaranteed solution.

7.2.1 MF-Site

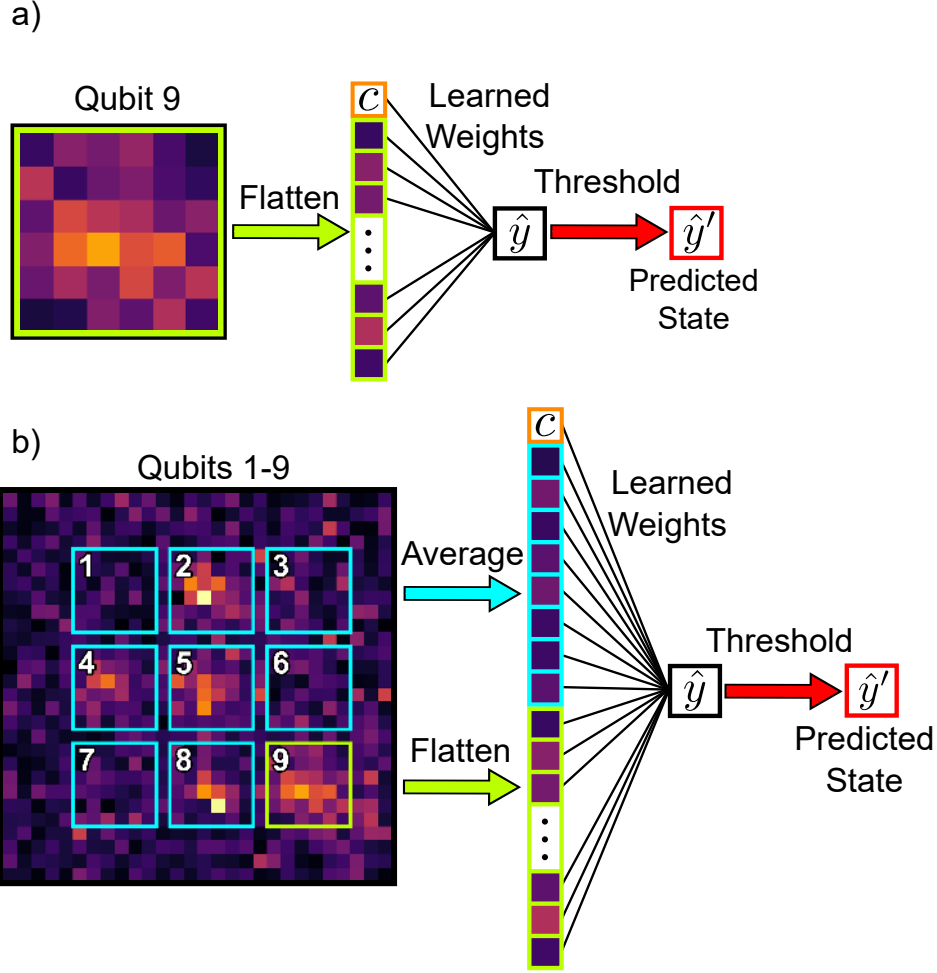


Figure 7.1: Matched filter designs for a single qubit with boundary size $s = 6$. (a) MF-Site configuration. (b) MF-Array configuration.

The MF-Site model classifies the state of a single qubit using only its corresponding pixel data, as illustrated in Fig. 7.1a for an example image where qubit 9 is in a bright state. The pixel data for this qubit is determined by first finding the center of each qubit in the image (see Appendix 8.7), and then placing a square boundary with width s . The pixel intensities I_{ij} inside each boundary are augmented by a constant c and placed in a flattened data structure known as the feature vector, depicted in Fig. 7.1a.

A dot product is formed by the feature vector and a vector containing the weights W_{ij} , which

gives the predicted state $\hat{y}$. Here, $\hat{y}$ is continuous but should be close to 0 for a dark state and 1 for a bright state. Last, a threshold is applied to $\hat{y}$ to produce a binary classification $\hat{y}' \in [0, 1]$, corresponding to the state estimate. Details about the numerical procedure for determining the weights and optimal threshold are given in Appendix 8.9.

7.2.2 MF-Array

The MF-Array model also classifies the state of a single qubit, but it also uses information from neighboring qubits to detect crosstalk, as depicted in Fig. 7.1b. In this example image, the goal is to classify the state of qubit 9 when qubits 1, 3, 6, and 7 are in the dark state, and qubits 2, 4, 5, 8, and 9 are in the bright state.

Boundaries are placed around each qubit and the pixels inside the boundary corresponding to the target qubit are flattened to form part of the feature vector, as in the MF-Site model. Pixels inside the boundaries around the other qubits are averaged and included in the feature vector, providing the model with information about the neighbor states. Since averaging primarily involves addition and introduces only a few additional values to the feature vector, the MF-Array has only a slight increase in computational complexity compared to the MF-Site model for a given boundary size. The weights and threshold are optimized similarly to the MF-Site model, with details given in Appendix 8.9.

Including the average pixel values from neighboring qubits allows the model to detect crosstalk, as a high average from a neighboring qubit suggests it is in the bright state, contributing more fluorescence that can reach the target qubit's pixels. By accounting for this effect, the model can better distinguish the target qubit's true state from crosstalk interference.

The advantages of this approach are that it can detect crosstalk with minimal computational overhead, and significantly reduces the amount of image data the model must process. Unlike other array-based methods that analyze entire pixel grids, this model condenses relevant information into just a few averaged values, capturing neighboring qubit influences without directly processing all surrounding pixels.

7.3 Results

In this section, I compare the classification performance, crosstalk mitigation, and discuss the physical insights provided by the learned weights of the matched filter. I also compare the computational complexity of the MF-Site and MF-Array to the traditional approaches and Phuttitarn's CNN algorithms [30].

7.3.1 Classification performance

Figure 7.2 shows the infidelity as a function of the qubit measurement time. The general trend is that infidelity decreases for longer measurement times as mentioned above. The goal is to reduce the measurement time for a given infidelity threshold.

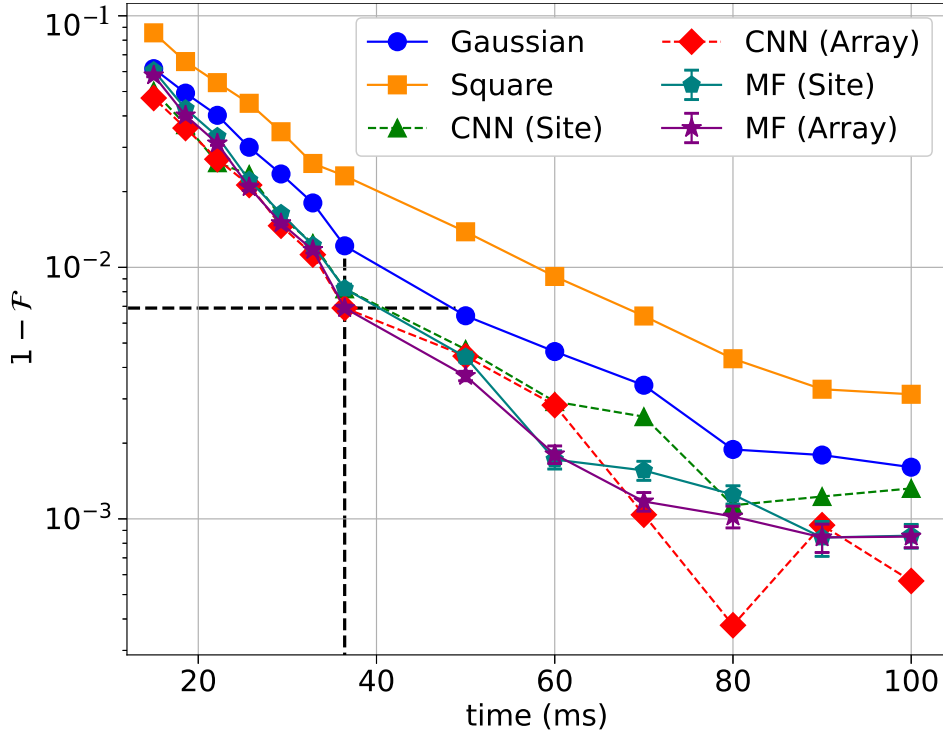


Figure 7.2: The measurement infidelity from different analysis methods at different readout times. The square, Gaussian, CNN (site) and CNN (array) infidelity values are taken from Ref. [30]. The error bars represent the standard error of shuffling the dataset with 10 random seeds.

For the traditional filters, we see that the square filter performs the worst (larger infidelity for any measurement time), whereas the Gaussian filter has a marked improvement because it accounts

for the expected intensity profile of the fluorescent light emitted by each qubit.

We see that all four classical machine learning algorithms substantially improve upon the traditional Gaussian filter algorithm. These algorithms perform equally well to within the apparent variation in the data, although the matched filter occasionally outperforms the CNN algorithms.

The error bars for the matched filter approaches represent the standard error, calculated from infidelity variations across 10 different random shuffles of the dataset. This ensures more representative fidelity curves, as single misclassifications at longer readout times, such as the CNN-Array at 80 ms, can cause large fidelity jumps that do not reflect overall performance. See Section 8.8 for details.

The vertical black dashed line indicates a measurement time of 36ms, at which Phuttitarn *et al.* [30] report maximum infidelity reductions for their CNN-site and CNN-array of 32% and 43%, respectively, relative to the Gaussian approach. At this measurement time, the MF-site and MF-array approaches have matching infidelity reductions of 32% and 43%, respectively.

The horizontal black dashed line indicates the point where Phuttitarn *et al.* [30] report the largest reduction in measurement time, with CNN-site and CNN-array achieving 20% and 25% reductions, respectively, relative to the Gaussian approach. At the same fidelity threshold, the MF-site and MF-array methods have matching readout time reductions of 20% and 25%, respectively.

We also see that the MF-Array model generally matches or outperforms the MF-Site approach, especially for longer readout times where crosstalk is more likely. This indicates that including the averages of neighboring qubit pixel intensities in the feature vector can improve performance.

Figure 7.3 shows the classification infidelity for each qubit using each approach at a measurement time of 36 ms. As before, the goal is to minimize the infidelity for each qubit, but the center qubit (site 5) is particularly important since it interacts with the most neighbors, serving as an indicator of how performance scales in larger qubit systems.

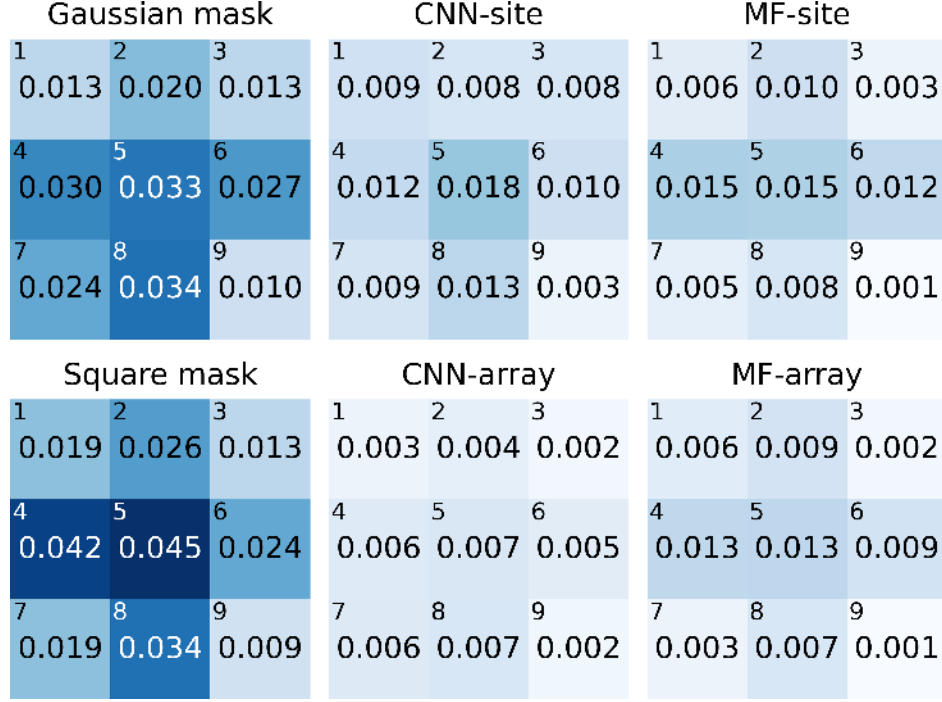


Figure 7.3: Classification infidelity for each qubit (numbered in top left of each box) for a measurement time of 36 ms. The standard error for the matched filter values is below 0.0001 for each site.

We see that both the Square and Gaussian approaches have significantly higher infidelity across all qubits compared to the other approaches, with the most significant increases being for the central qubit and its nearest neighbors. This highlights that the traditional approaches cannot account for crosstalk.

Looking at the other approaches, we see that CNN-Site, MF-Site, and MF-Array perform similarly well, all improving center qubit performance over traditional methods, with MF-Array achieving the lowest infidelity of the three. The CNN-Array further reduces the infidelity for the center qubit and its nearest neighbors but comes with a significant increase in computational complexity compared to the CNN-Site, as discussed later.

Despite the CNN-Array providing better crosstalk reduction than the MF approaches, the MF-Site and MF-Array both provide significant infidelity reductions across all qubits compared to the traditional approaches, and offer comparable performance to the CNN-site.

Table 6 shows the cross-fidelity (Eq. 22) for the center qubit to its nearest neighbors (C-NN)

and for the edge sites (E-E) at a measurement time of 36 ms. Positive values for a given qubit pair indicate that the predictions of their states are correlated, while negative values signify anti-correlation. Ideally, the cross-fidelity magnitude should be close to zero for a given qubit pair, indicating that the model is mitigating the effects of crosstalk.

Table 6: Comparison of cross-fidelity values at a measurement time of 36 ms for different methods. The mean standard errors for the MF-Site and MF-Array are 0.0004 for the C-NN values and 0.0002 for the E-E values. The boundary size s is optimized for both the MF-Site and MF-Array.

C-NN		Gaussian [30]	Square [30]	CNN-site [30]	CNN-array [30]	MF-Site	MF-Array
1	$\mathcal{F}_{52}^{\text{CF}}$	0.0298	0.0524	0.0151	0.0099	0.0108	0.0102
2	$\mathcal{F}_{54}^{\text{CF}}$	0.0211	0.0298	0.0146	0.0099	0.0058	0.0066
3	$\mathcal{F}_{56}^{\text{CF}}$	0.0218	0.0284	0.0120	0.0022	0.0071	0.0074
4	$\mathcal{F}_{58}^{\text{CF}}$	0.0392	0.0607	0.0120	0.0020	0.0096	0.0077
5	$\langle \mathcal{F}_{5l}^{\text{CF}} \rangle$	0.0280	0.0428	0.0134	0.0060	0.0083	0.0080
E-E		Gaussian [30]	Square [30]	CNN-site [30]	CNN-array [30]	MF-Site	MF-Array
6	$\mathcal{F}_{13}^{\text{CF}}$	0.0110	0.0139	0.0205	0.0110	0.0104	0.0090
7	$\mathcal{F}_{79}^{\text{CF}}$	-0.0036	-0.0005	-0.0094	-0.0036	-0.0063	-0.0090
8	$\mathcal{F}_{17}^{\text{CF}}$	0.0038	0.0091	-0.0016	0.0038	-0.0006	-0.0027
9	$\mathcal{F}_{39}^{\text{CF}}$	0.0028	0.0090	0.0031	0.0028	0.0002	-0.0009
10	$\mathcal{F}_{19}^{\text{CF}}$	-0.0068	-0.0043	-0.0098	-0.0068	-0.0077	-0.0089
11	$\mathcal{F}_{37}^{\text{CF}}$	0.0070	0.0078	0.0080	0.0070	0.0063	0.0056
12	$\langle \mathcal{F}_{kl}^{\text{CF}} \rangle$	0.0058	0.0074	0.0087	0.0058	0.0052	0.0060

We see that for the nearest neighbors, both the MF-Site and MF-Array approaches achieve average cross-fidelity magnitudes over three times lower than those of the Gaussian approach, and lower than the CNN-Site approach. While the CNN-Array has the lowest cross-fidelity for the nearest neighbors, the MF-Array is not significantly higher and shows an improvement over the MF-Site.

Looking at the edge sites, we see that the cross-fidelity magnitude is typically much smaller compared to the magnitude of the nearest neighbors for most approaches. This aligns with expectations as edge sites are spaced further apart, reducing the likelihood of crosstalk. While cross-fidelity varies between approaches, the overall lower crosstalk at edge sites makes these differences negligible.

Given these cross-fidelity metrics, I conclude that for nearest neighbors, both the MF-Site and MF-Array significantly reduce crosstalk compared to the standard approaches while achieving

performance comparable to the CNN methods. Notably, the additional features in MF-Array that incorporate information about neighboring qubit states further reduce crosstalk compared to MF-Site.

7.3.2 Physical interpretability of weights

One strength of the matched filter approach is that the learned weights provide physical insights into the optimal filter. Figure 7.4 shows the weights for each qubit in the MF-Site model with a boundary size of $s = 10$ at a measurement time of 36 ms. The i and j axes represent the pixel coordinates, and the vertical scale represents the weight values W_{ij} . At this boundary size, qubit regions overlap, allowing the model to account for crosstalk to some degree.

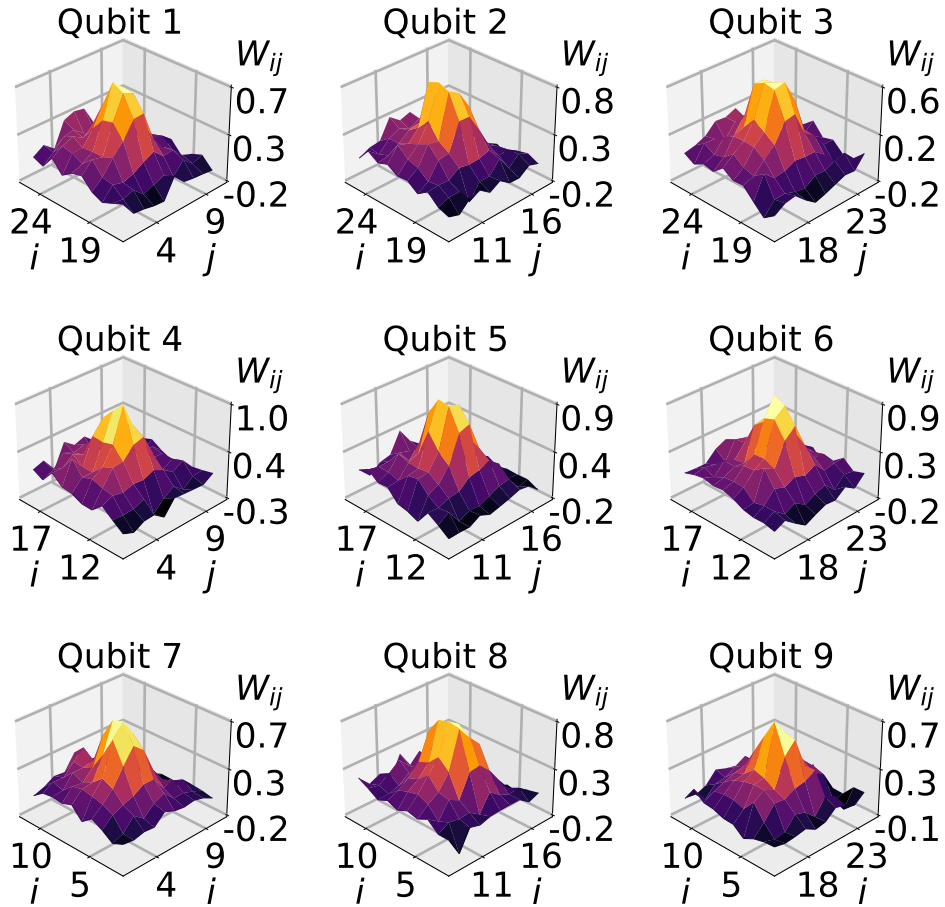


Figure 7.4: False-3D images of the learned weights for each qubit.

We see that the weights have a Gaussian-like shape, but the weights become increasingly

bumpy as the distance from the center increases for some qubits. I hypothesize that the optimal filter weights deviate from a Gaussian to account for imperfections in the experimental setup, such as optical aberrations or pixel quantum efficiency variation.

The variation in the weights near the boundary of each region is affected by the least-squares regularization parameter α . Optimizing α has only a minor effect on model prediction accuracy, so I set it equal to zero (normal least-squares optimization) for simplicity. Increasing α imposes a penalty on large weights, which smooths out the bumps and produces a more Gaussian-like shape.

We can also visualize the weights that correspond to the neighboring qubits in the MF-Array approach, shown in Figure 7.5 for the optimal boundary size $s = 8$ at a readout time of 36 ms. The general trend is that the nearest neighbor qubits, particularly those below and to the right of each qubit, exhibit large negative weight values, indicating that the model is accounting for crosstalk from those qubits. In contrast, the qubits above and to the left of the target qubit often show small, near-zero positive weights. This disparity may reflect imperfections in the optical setup, causing the fluorescence from neighboring qubits to propagate in specific directions. The additional physical interpretability provided by the MF-Array may help to identify and address such setup-related imperfections.

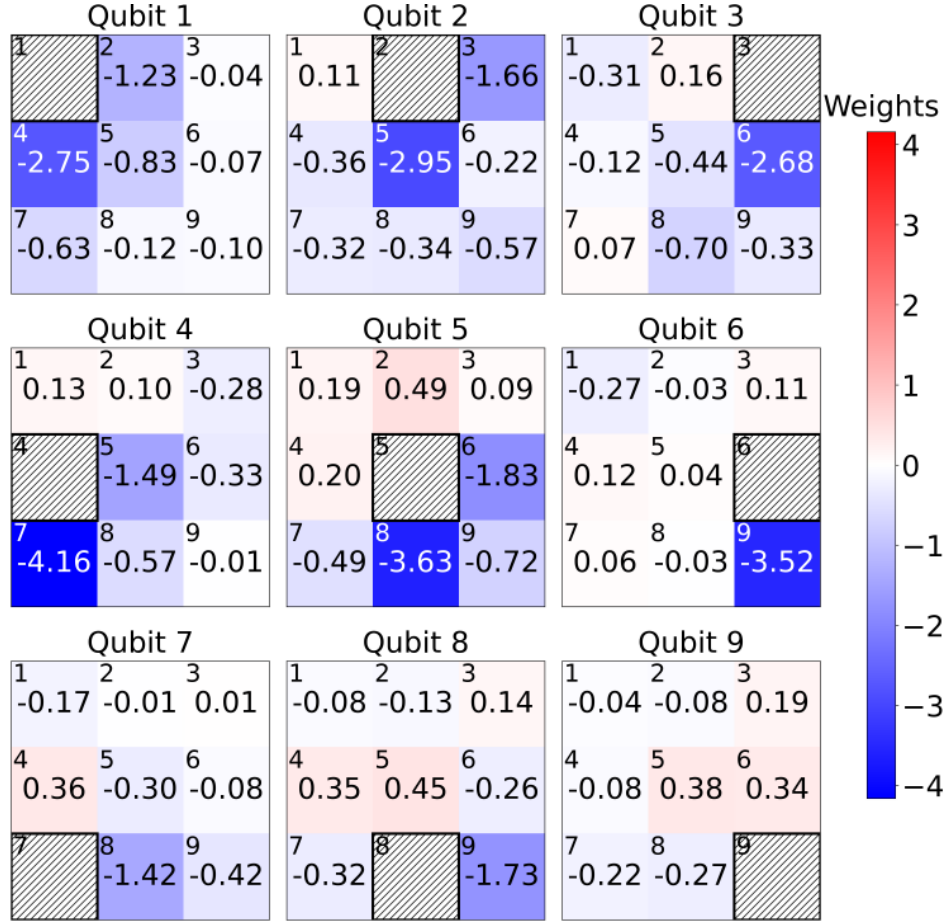


Figure 7.5: Weights of neighbor qubits (numbered in the top left of each box) for the MF-Array model.

7.3.3 Algorithm computational complexity

Next, I compare the computational complexity of the approaches by considering the number of trainable parameters for each model and the number of multiplications and nonlinear functions required during evaluation. Table 7 summarizes these factors and also includes the mean classification fidelity across all qubits and measurement times for reference.

Table 7: Comparison of the number of trainable parameters, evaluation complexity, and fidelity of different approaches. The standard error of the mean fidelity is 0.0003 for both the MF-Site and MF-Array.

Model	# Trainable Param.	# Mult.	# Nonlinear Functions	Mean Fidelity
Square	0	0	0	0.9712
Gaussian	18	855 (2.5×10^5)	0	0.9804
CNN-Site	3.6×10^5	1.9×10^7	5.9×10^4	0.9856
CNN-Array	7.5×10^7	1.2×10^8	1.2×10^5	0.9866
MF-Site ($s = opt.$)	909–1731	909–1731	0	0.9841
MF-Array ($s = opt.$)	522–1576	522–1576	0	0.9851

We see that the traditional square filter approach requires no trainable parameters, multiplications, or nonlinear functions, making it the simplest to evaluate but it results in the lowest mean fidelity. The Gaussian approach has 18 trainable parameters which correspond to the standard deviation and amplitude of the Gaussian fit for each qubit. As mentioned before, in Ref. [30] the Gaussian filter for each qubit spans the entirety of the raw 442×62 -pixel image, bringing the total number of multiplications to 2.5×10^5 . However, this is not practical for real-time evaluation. Instead, I estimate the number of multiplications for the Gaussian filter by selecting only the values in the filter greater than 10^{-3} within its 0 to 1 range. Despite their lower performance, these traditional approaches scale linearly with the number of qubits, making them practical for systems with a large number of qubits.

In contrast, the CNN-site and CNN-array approaches require many orders of magnitude more trainable parameters, multiplications, and nonlinear functions compared to the traditional approaches, but offer significant increases in fidelity. While the CNN-site model scales linearly with the number of qubits, evaluating it for even a single qubit can exceed the resources of a single chip. In its current form, the CNN-array scales non-linearly with the number of qubits because the input image size must increase to include the additional qubits. This may not require adjustments to the number or shape of the convolutional layers, but it would greatly increase the number of neurons in the final layers of the model. The CNN-array model could be modified to predict only the center qubit state while keeping the same input pixel size, but this would still require a substantial computational load during both training and evaluation.

The complexity metrics for the matched filter approaches are presented as ranges, as the boundary size s is optimized for each qubit and measurement time (see Appendix 8.10). In some cases, the MF-Array uses fewer trainable parameters and multiplications than the MF-Site, despite the MF-Array’s ability to process multiple regions of the image. This is likely because the optimal MF-Site tends to require larger boundary sizes s to capture information from neighboring qubits, whereas the MF-Array inherently incorporates this information by design.

The matched filters require a larger number of parameters than the traditional approaches, but two (four) orders of magnitude fewer than the CNN-Site (CNN-Array) approaches. Moreover, the linear optimization used to determine these parameters is significantly faster and requires fewer computational resources.

The number of multiplications required for the matched filter approaches is slightly larger on average than the Gaussian filter, but this is only because the optimal boundaries include more pixels. In some cases, the optimal MF-Array uses less multiplications than the Gaussian. Like the Gaussian approach, the matched filters do not require the evaluation of any nonlinear functions. Despite the matched filters requiring four (five) orders of magnitude fewer multiplications and nonlinear function evaluations than the CNN-Site (CNN-Array) approaches, they achieve similar mean fidelities.

Critically, the number of operations in the MF-Site approach scales linearly with the number of qubits, and in the MF-Array approach, limiting the feature vector to include only nearest neighbors also results in linear scaling. In addition, averages can be converted to sums, with divisions absorbed into the weights, further reducing the number of operations.

7.4 Discussion

My algorithms require a set of multiplications that scale linearly with the number of qubits and hence can be applied to large qubit arrays. Furthermore, the multiplications can be done in parallel using, for example, a field-programmable gate array, where the computations can be executed in a few clock cycles. This will allow the qubit state readout computations to be completed on the

nanosecond time scale, which is negligible compared to the measurement time.

Learning the matched filter requires only regularized least-square regression, which can be done one feature vector at a time using sequential regression methods [71]. This will allow for on-chip learning without storing the dataset or transferring it to another computer for processing. Constantly updating the weights will allow the matched filter to follow changes in the experimental setup.

Chapter 8

Conclusions and Future Research

In this chapter, I start by discussing the results and their potential impact on future quantum computers. I then discuss future research directions.

8.1 Discussion

In this thesis, I demonstrate that a next-generation reservoir computer (NG-RC) possesses many of the key properties necessary for fast and accurate qubit state discrimination for two leading quantum computing platforms: superconducting qubits and neutral atom qubits. This includes an efficient training procedure that enables real-time updates to adapt to changing experimental conditions and a computational complexity that scales favorably with the number of qubits, making it suitable for the larger quantum computers of the future.

For superconducting qubit systems, I perform time-averaging on the input signals and system identification to reduce the number of terms in the model, and explore using different polynomial degrees for the nonlinear features. After developing these NG-RC models, I tested them on both single-qubit and 5-qubit datasets.

For all datasets tested, at least one NG-RC variant achieved higher measurement fidelity than traditional approaches while requiring similar or fewer multiplications during evaluation, and in some cases, up to two orders of magnitude fewer multiplications than the largest feedforward neural network. As the number of multiplications directly correlates to the computational complexity of the model, these results indicate that the NG-RC is suitable for fast and high-fidelity superconducting qubit state discrimination. Furthermore, the NG-RC models with linear terms scale linearly in complexity as the number of the system increases, making this approach scalable for large quantum computers of the future. Finally, I demonstrate that the learning can be done in batches or sequentially, allowing learning with large datasets or real-time updates when evaluated

on control hardware.

For neutral atom qubit systems, I develop two NG-RC models. The first uses pixel values from fluorescence imaging as inputs to the model, while the second also uses average pixel values from neighboring qubits to detect crosstalk. I test both of these approaches on a 3×3 array of neutral atoms.

Both NG-RC variants achieve similar fidelity to a convolutional neural network (CNN) that has two (four) orders of magnitude more trainable parameters (number of multiplications). In addition, the NG-RC model that accounts for neighboring qubits shows better crosstalk reduction than one of the CNNs. Furthermore, if the number of neighbors in the crosstalk model are limited, both NG-RC models scale linearly in complexity as the number of qubits increases. These results indicate that an NG-RC is a viable solution for fast, accurate, and scalable neutral atom quantum computers of the future.

In addition, I demonstrate that the learned weights of the NG-RC are physically interpretable, revealing potential imperfections in the experimental setup, such as optical aberrations and the direction of misdirected fluorescent light in the crosstalk model. These qualities have the potential to diagnose experimental problems in current and future neutral atom systems.

8.2 Future Directions

The results presented in this thesis lay the groundwork for several promising directions in future research.

For superconducting qubit systems, a natural next step is to implement NG-RC models directly on control hardware, such as field-programmable gate arrays (FPGAs), to process incoming signals in real time. NG-RC models require significantly fewer hardware resources compared to neural network implementations and could achieve evaluation times in a few nanoseconds, leaving additional resources available for tasks such as qubit control, state discrimination, or real-time learning. Sequential learning could be implemented in the FPGA fabric or on an integrated processor to enable continuous weight updates as new data is acquired. Integration with platforms like the

Quantum Instrumentation Control Kit (QICK) [72], an open-source framework for RFSoc-based FPGAs with Python programmability, could further simplify deployment and make these models more accessible for experimental use.

To further the goal of the MURI collaboration (see Section 1.1), the NG-RC could be used as the classical backend after multiple quantum processing layers to realize a quantum reservoir processor.

For neutral atom systems, the NG-RC could be tested on larger neutral atom arrays, which would highlight its scalability compared to current CNNs. In addition, it could be implemented on a computer chip integrated in the camera for rapid state discrimination. Last, it would be interesting to test the performance of an NG-RC on non-destructive measurements, which are crucial in error correction schemes.

In this thesis, I show that there are different ways to configure the NG-RC, such as time-averaging or polynomial order, but there may be other variations that further increase performance or simplify the model. One option is to pre-process the data using matched filters as in Ref. [24] before applying nonlinearity to keep the complexity low. Another option is to use nonlinear functions other than polynomials, such as nonlinear kernels.

In summary, NG-RCs are promising for fast, high-fidelity, and scalable qubit state discrimination, with many avenues for future research.

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Appendix A: Methods for Superconducting Qubit Readout

8.3 Feature Creation

For the single-qubit datasets (Figure 5.2a-c), the number of time steps (or $\mathcal{I}\mathcal{Q}$ pairs) in each readout measurement is 200, 200, and 202, respectively.

To reduce the number of features, data is averaged within non-overlapping time windows of fixed sizes $w = 1, 2, 5, 10, 25, 50$, and 100 for the linear NG-RC; 20, 25, 50, and 100 for the quadratic NG-RC; 25, 50, and 100 for the cubic NG-RC; 5, 10, 20, 25, 50, 100 for the linear NG-RC with system identification; 20, 25, 50, 100 for the quadratic NG-RC with system identification; and 25, 50, and 100 for the cubic NG-RC with system identification. For the system identification results, only the window size that maximizes the fidelity is shown for the linear, quadratic, and cubic models in Figure 5.2a-c, for readability.

For each dataset and each set of measurement conditions, 50% of the data is used for training the models, and 50% of the data is used for testing the models. The quadratic NG-RCs contain a constant, the linear features and a constant, and the quadratic monomials. The cubic NG-RCs augments the quadratic model by including the cubic monomials.

For the multi-qubit datasets, the readout signal, which originally contained 500 time steps, was digitally demodulated. Next, a boxcar filter Eq. 13 is applied to the individual qubit signals. Starting with the optimal boxcars from Lienhard *et al.* [20], I shift each boxcar by a single time step until the testing fidelity is maximized. The resulting boxcars begin at the first time step and end at time steps 500, 500, 282, 479, and 295 for the five qubits, respectively. In contrast, no boxcar filters are applied for the linear NG-RC that uses the raw, non-demodulated signals.

After applying the boxcars, the individual signals are averaged with $w = 10, 50, 100, 200, 250$, and 500 for the linear NG-RC; 50, 100, 200, 250, and 500 for the quadratic NG-RC; and 200, 250, and 500 for the cubic NG-RC. For all experiments, only a single value of w is used for each model,

but a combination of window sizes could be used to maximize performance. In the multi-qubit case, an individual NG-RC is trained to predict the state of a single qubit while having access to features from the other qubit signals.

8.4 System Identification

To perform the system identification on the single qubit datasets, I use the package `SysIdentPy` [47] with the "NFIR" model type, "ridge_regression" estimator, and the Polynomial basis function. I perform a feature search for multiple values of `ridge_param`, as discussed in the next subsection. For each NG-RC type, I truncate `n_terms` when the difference between successive information values drops below 1% of the maximum range to ensure the model remains compact. For the linear NG-RC, I use `degree=1` and the maximum `n_info_values` for the information criteria search. For both quadratic (`degree=2`) and cubic (`degree=3`) NG-RC, I use `n_info_values=30` to balance performance and evaluation time.

8.5 Single-Qubit Model: Training and Testing

During the training phase, the feature vectors $\mathcal{O}_{total,m} \in \mathbb{R}^{N_f \times 1}$ from each measurement m in the training dataset are concatenated to form the matrix $\mathcal{O}_{total} \in \mathbb{R}^{N_f \times M}$, where N_f is the length of each feature vector and M is the total number of measurements. Similarly, the ground truth labels for each measurement $y_m \in \mathbb{R}^{d \times 1}$ in the training dataset are concatenated to form the matrix $\mathbf{Y} \in \mathbb{R}^{d \times M}$, where the dimension d of the output depends on whether one-hot encoding or threshold-based outputs are used. The weights W_{out} are computed using regularized linear regression with an L2 regularization term, known as ridge regression [73], given by

$$W_{out} = \mathbf{Y} \mathcal{O}_{total}^\top (\mathcal{O}_{total} \mathcal{O}_{total}^\top + \alpha \mathbf{I})^{-1} \quad (24)$$

where α is the regularization parameter. Larger values of α penalize the size of the weights, and ensures that $\mathcal{O}_{total} \mathcal{O}_{total}^\top$ is not ill-conditioned.

For each of the single qubit datasets and measurement conditions, I compute Equation 24 for all measurements m in the training dataset, with α values logarithmically spaced between 10^{-7} to 10^{-1} , as well as $\alpha = 0.0$. I then evaluate the performance of each of these weights by predicting $Y = W_{out}\mathcal{O}_{total}$ for each α , where Y contains the prediction for every measurement m in the testing dataset. If the model has a single output (rather than one-hot encoding), a threshold is applied to determine the predicted state. This threshold is searched within the range $[-0.5, 0.5]$ in steps of 0.01. The combination of α and threshold that achieve the highest fidelity on the testing dataset is selected. If the output is one-hot encoded, predictions are made by choosing the index of y_m that has the highest value. In that case, the α value is chosen that maximizes the fidelity on the testing set, without performing a threshold search.

8.6 Multi-Qubit Model: Training and Testing

Due to the significantly larger size of the multi-qubit datasets, the matrix multiplications and inversions in Eq. 24 are too costly to evaluate using the entire training dataset at once. Therefore, I perform the training in batches by using a variation of sequential ridge regression originally described by [71].

I begin with the solution to ordinary least squares, which is equivalent to Eq. 24 for $\alpha = 0$, given by

$$W_{out} = \mathbf{Y}\mathcal{O}_{total}^\top (\mathcal{O}_{total}\mathcal{O}_{total}^\top)^{-1}, \quad (25)$$

and I express the matrix multiplications as sums by writing

$$W_{out} = \sum_{l=1}^K \tilde{Y}_l \tilde{\mathcal{O}}_l^\top \left(\sum_{l=1}^K \tilde{\mathcal{O}}_l \tilde{\mathcal{O}}_l^\top \right)^{-1}. \quad (26)$$

Here, $\tilde{Y}_l \in \mathbb{R}^{d \times k}$, k is number of measurements in each batch, K is the total number of batches,

and $\tilde{O}_l \in \mathbb{R}^{k \times N_f}$ is the feature vector for batch l . These sums are equivalent to iterating the maps

$$Q_{l+1} = Q_l + \tilde{Y}_{l+1} \tilde{O}_{l+1}^T \in (d \times N_f), \quad (27)$$

$$P_{l+1} = P_l + \tilde{O}_{l+1} \tilde{O}_{l+1}^T \in (N_f \times N_f), \quad (28)$$

$$W_{\text{out},l+1} = Q_{l+1} (P_{l+1})^{-1}. \quad (29)$$

In this iterative approach, the regularization parameter is applied through the initial conditions by taking

$$Q_0 = 0, \quad (30)$$

$$P_0 = \alpha \mathbf{I} \in (N_f \times N_f). \quad (31)$$

We see that this approach requires the inverse of an $(N_f \times N_f)$ matrix, which can become inefficient when the number of features is large. However, this approach is increasingly efficient when the batch size $k \gg N_f$.

The original approach by Hertz [71] updates the weights using a single measurement ($k = 1$), which eliminates the need for matrix inversion entirely by applying the Sherman-Morrison Identity [74]. This approach is likely the most efficient option for training a discriminator in real time on an FPGA. However, processing multiple measurements simultaneously is often more computationally efficient than handling them one at a time when performing the training in Python on a desktop computer.

Another alternative is to use the Woodbury matrix identity [75] to modify the mapping to require an inverse of a $k \times k$ matrix, which is inefficient for a large number of measurements in each batch. In this work, I take $k = 32,000$ and the number of features is typically less than a few thousand, making Eq. 29 the most efficient in my case.

The training set contains 15,000 measurements per possible state so that there are 15 total batches used to train the model. In the first batch, I initialize Eqs. 31 for regularization parameters

logarithmically spaced between 10^{-7} to 10^3 , as well as $\alpha = 0.0$. I iterate through each batch, computing Eq. 29 for each α , and save each copy of the weights for the testing phase.

I also perform the testing phase in batches by computing $Y = W_{out}\mathcal{O}_{total}$ for k measurements in each batch. I select the optimal threshold and α for each qubit that maximizes the classification fidelity over the entire testing set.

Appendix B: Methods for Neutral Qubit Readout

8.7 Preprocessing

The first step is to crop the full 442×62 -pixel image to obtain the 28×28 -pixel image corresponding to the secondary path. For each measurement time, there are 6,002 images, randomly shuffled and split into 60% for training, 20% for testing, and 20% for validation. Next, I find the mean pixel intensity across all training images and subtract it from all images. Finally, I determine the maximum range of intensities across all images and normalize the data by this value.

For all algorithms, I must determine the location of each qubit in the image. This is accomplished by averaging all images in the training dataset, which produces an image with bright spots in the location of each qubit. I use the `peak_local_max` function from the `skimage.feature` package to determine an estimate of the bright spots in the image, and then use these coordinates as initial guesses for Gaussian fits using `curve_fit` from the `scipy.optimize` package. The fitted coordinates are used as the center for the boundaries for each qubit.

8.8 Estimating Statistical Uncertainty in Infidelity

As mentioned in the previous section, the images are split and randomly shuffled into training, testing, and validation sets. For Figures 7.2 and 7.3, and Tables 6 and 7, the dataset is split using 10 independent random shuffles, the infidelities at each measurement time are computed, and the infidelities for each random shuffle are averaged. The error bars and quoted uncertainties represent the standard error, or the standard deviation of the values over the 10 random shuffles, and then divided by $\sqrt{10}$. These uncertainties do not account for any systematic uncertainties present in the experiment.

8.9 Training

In the training step, the feature vectors for each image $\mathbf{x} \in \mathbb{R}^{d \times 1}$ (see Fig. 7.1), where d is the number of elements in the feature vector, are placed into a matrix $\mathbf{X} \in \mathbb{R}^{d \times M}$, where M is the number of images in the training dataset, and the corresponding ground truth labels from the primary path $y \in [0, 1]$ are placed into a matrix $\mathbf{Y} \in \mathbb{R}^{1 \times M}$.

The weights $\hat{\mathbf{W}}$ are learned using linear regression with an L2 regularization, also known as Tikhonov regularization [73], which has a guaranteed solution given by

$$\hat{\mathbf{W}} = \mathbf{Y}\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \alpha \mathbf{I})^{-1} \quad (32)$$

where α is the regularization parameter. Regularization penalizes the size of the weights and compensates for possible ill-conditioning of the matrix $\mathbf{X}\mathbf{X}^\top$.

I evaluate Eq. 32 for each qubit and measurement time, setting $\alpha = 0$, and store the corresponding weights. I also compute the weights for different boundary sizes s ranging from 2 to 14 in steps of 1. The optimization the threshold and the boundary size are detailed in the next section.

8.10 Optimizing model metaparameters

For each qubit and measurement time, I simultaneously optimize the threshold and boundary size. For each boundary size, the weights learned in the training step are used to predict the qubit states for all feature vectors $\mathbf{x}$ in the validation set by computing $\hat{y} = \hat{\mathbf{W}}\mathbf{x}$. Next, $\hat{y}'$ is computed for all thresholds ranging from 0.01 to 0.99 in steps of 0.01. Lastly, the combination of boundary size and threshold are chosen that maximize the classification fidelity on the validation set.

8.11 Testing

Lastly, the optimal boundary size, weights, and thresholds for each qubit are used to obtain $\hat{y}'$ for each image in the testing image set and compared to the ground truth values y to obtain the

classification fidelity.

8.12 Cross-Fidelity Calculation

To obtain a more representative cross-fidelity calculation, I apply the trained models to a larger dataset of 39,976 images instead of the typical 1,200-image test sets. This is because the expanded dataset includes more examples of the possible permutations of the 9-qubit states, allowing for a more comprehensive assessment of crosstalk effects and improving the reliability of the reported cross-fidelity metrics.